

The mental Healthcare intelligent Virtual Reality solutions consortium
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Me-Health innoVatoRs

1. General information

1.a Applicants

In case an applicant has more than one affiliation, please mention this in the table.

Name*	Institution	Department	Email	Tenured (i.e. fixed) position/	For Erasmus MC applicants: clinician/ researcher/ mix	Role in Flagship [Main Lead/Lea d/
				Tenure- track/ Other (please specify)		Applicant/ Co- applicant]
Prof.dr. Matthias Wieser	Erasmus University Rotterdam	Clinical Psychology	wieser@essb.eur.nl	Tenured position	n.a.	Main Lead
Dr. Janne Papma	Erasmus MC	Neurology	j.papma@erasmusmc.nl	Tenured position	Researcher	Lead
Prof.dr. Bart De Schutter	Delft University of Technology	Delft Center for Systems and Control	B.deschutter@tudelft.nl	Tenured position	n.a.	Lead
Dr. Marta Andreatta	Erasmus University Rotterdam	Psychology, Education and Child Studies	andreatta@essb.eur.nl	Tenured position	n.a.	Applicant
Dr. Ruth Van der Hallen	Erasmus University Rotterdam	Psychology, Education and Child Studies	vanderhallen@essb.eur.nl	Tenured position	n.a.	Applicant
Prof. dr. Yvonne van Everdingen	Erasmus University Rotterdam	Marketing Management	yeverdingen@rsm.nl	Tenured position	n.a.	Applicant

Dr. Janna Cousijn	Erasmus University Rotterdam	Clinical Psychology	cousijn@essb.eur.nl	Tenure-track	n.a.	Applicant
Dr. Harro Seelaar	Erasmus MC	Neurology	h.seelaar@erasmusmc.nl	Tenured position	Clinician and researcher	Applicant
Dr. Esther van den Berg	Erasmus MC	Neurology	e.vandenberg@erasmusmc.nl	Tenured position	Clinician and researcher	Applicant
Dr. Jackie Poos	Erasmus MC	Neurology	J.m.poos@erasmusmc.nl	Post-doc, to be hired	Researcher	Applicant
Dr. Anahita Jamshidnejad	Delft University of Technology	Control and Operations	A.Jamshidnejad@tudelft.nl	Tenured position	n.a.	Applicant
Prof. dr. Mark Neerincx	Delft University of Technology	Interactive Intelligence	M.A.Neerincx@tudelft.nl	Tenured position	n.a.	Applicant
Dr. ir. Iulia Lefter	Delft University of Technology	Multi-Actor Systems	i.lefter@tudelft.nl	Tenured position	n.a.	Applicant
Dr. ir. Rafael Bidarra	Delft University of Technology	Intelligent Systems Department - Computer Graphics and Visualization	R.Bidarra@tudelft.nl	Tenured position	n.a.	Applicant
Dr. Michael Weinmann	Delft University of Technology	Intelligent Systems Department - Computer Graphics and Visualization (CGV)	M.Weinmann@tudelft.nl	Tenure-track	n.a.	Applicant
Dr. Ricardo Marroquim	Delft University of Technology	Intelligent Systems Department - CGV	R.Marroquim@tudelft.nl	Tenured position	n.a.	Applicant
Prof. dr. Elmar Eisemann	Delft University of Technology	Intelligent Systems Department - CGV	E.Eisemann@tudelft.nl	Tenured position	n.a.	Applicant

1.b Summary (max 300 words) word count: 299

One in every two people worldwide experiences a mental illness, which significantly impacts individuals, families, and communities in terms of quality of life and socio-economic burden¹. The corresponding costs in Europe are estimated over €600 billion². Reliable and timely diagnosis and treatment are expected to significantly reduce these costs. However, currently clinical evaluations are paper-and-pencil-based and subjective, which fail to encapsulate the richness and complexity of real-life situations, and consider a “one-size-fits-all” principle rather than personalized methods that fit the individual characteristics and needs of patients. Virtual reality (VR) is proven to have the potential to provide solutions that address these challenges^{3,4}. Such solutions, however, are not yet developed systematically nor are adopted by end-users (patients/clinicians). Our aim is to improve early diagnosis (WP1) and treatment (WP2) across mental disorders by feasible and acceptable (WP3) intelligent VR-based solutions. With *Me-Health InnoVatoRs*, we will establish a sustainable collaboration that will result in a national expertise center focused on intelligent VR-based solutions in mental healthcare. In the first five years, we will develop innovative VR-based solutions and the corresponding scientific groundwork for personalized diagnosis and treatment of frontotemporal dementia (FTD) and social anxiety disorder (SAD). Considering co-creation approaches, we will implement and generalize these solutions to provide foundations for long-term sustainability of *Me-Health innoVatoRs*. In the second five years, we will adopt and further develop our solutions for other mental disorders (e.g., addiction, post-traumatic stress disorder, autism), and for at-home VR-based treatment. *Me-Health innoVatoRs* will significantly improve the well-being and quality of life patients, families and communities by enabling earlier diagnosis and by increasing the success rates and accessibility of effective treatment methods. Consequently, the core costs for societies will significantly decrease, due to a reduction of patients entering specialized mental healthcare, shorter waiting lists, and a reduced burden on clinicians.

1.c Public Summary (max 300 words) word count: 300

Één op de twee mensen ontwikkelt een psychische aandoening gedurende het leven. Dit heeft ernstige gevolgen voor de kwaliteit van leven en kosten van patiënten, families en de maatschappij. De totale kosten in Europa worden geschat op meer dan €600 biljoen. Betrouwbare en vroege diagnostiek en behandeling kunnen deze kosten aanzienlijk verlagen, maar schieten nu nog tekort. Behandelaars zijn op dit moment afhankelijk van pen-en-papier tests en subjectieve vragenlijsten die niet goed de klachten van de patiënt in het dagelijks leven weergeven. Daarnaast zijn de huidige instrumenten niet goed afgestemd op de eigenschappen en wensen van de individuele patiënt. Met virtual reality (VR) technologieën kunnen wij het echte leven te benaderen, en VR biedt de mogelijkheid om diagnostiek en behandeling te personaliseren. Echter, wordt deze technologie op dit moment nog niet geaccepteerd en gebruikt door behandelaars en patiënten. Ons doel is om de vroegdiagnostiek en behandeling van psychische aandoeningen te verbeteren door deze effectiever en gepersonaliseerd te maken met behulp van VR technologie. Om dit te bereiken zullen wij een nationaal expertisecentrum op het gebied van ‘slimme’ VR technologie in de mentale gezondheidszorg opzetten: het *Me-Health InnoVatoRs* consortium. Binnen dit consortium zullen wij duurzame samenwerkingsverbanden met het bedrijfsleven, de geestelijke gezondheidszorg en patiëntenorganisaties opzetten. In de

eerste vijf jaar zullen wij ons specifiek richten op patiënten met dementie en patiënten met een angststoornis. Op de lange termijn, zullen wij dit uitbreiden naar andere psychische aandoeningen (bijv. verslaving, post-traumatische stress stoornis, autisme), en zullen wij VR behandelingen ontwikkelen die thuis gebruikt kunnen worden. Deze initiatieven zullen ervoor zorgen dat diagnoses eerder gesteld kunnen worden, behandelingen effectiever worden, en dat wachtlijsten voor behandeling zullen afnemen. Dit zal ervoor zorgen dat de kwaliteit van leven van patiënten en families verbeterd, de werkdruk voor behandelaars afneemt, en de kosten voor de maatschappij aanzienlijk afnemen.

2. Relevance to Convergence Health & Technology mission (50%) – max. 2500 words for 2.a to 2.e (word count: 2429)

2.a Contribution to H&T mission and fit with H&T theme(s) word count: 343 words

In this flagship we set the stage for an expertise center for intelligent advanced virtual reality (VR) solutions in mental healthcare, *Me-Health InnoVatoRs*. Our public-private consortium of scientific, societal, and industry partners aims to develop **personalized diagnostic and treatment VR solutions for mental health disorders**, developing innovative VR diagnostic (WP1) and treatment (WP2) solutions for frontotemporal dementia (FTD) and social anxiety disorder (SAD). For this purpose, we follow interdisciplinary and iterative co-creation approaches to involve end-users throughout the entire VR development process, with a specific focus on eventual acceptance and adoption (WP3).

Consequently, *Me-Health InnoVatoRs* will contribute to all four central tasks of the H&T mission: 1) lifelong health, 2) socio-economic equality in healthcare, 3) early diagnostics to proactively maintain health, and 4) individually tailored precise medical treatment for all. The consortium explores the potentials of VR technologies for **individually tailored** health applications from adolescence until older age (promoting lifelong health). WP1 will improve **early diagnosis** in FTD and SAD (early diagnostics to proactively maintain health), whilst WP2 will focus on **personalized therapy** for SAD from adolescence onwards, and on relieving burden in early FTD stages (individually tailored precise medical treatment). Furthermore, to address social inequalities and proactively contribute to socio-economic equality in healthcare, the consortium will focus on people with **diverse (cultural) backgrounds** by making use of the knowledge and expertise of the cross-cultural outpatient memory clinic Erasmus MC, thereby contributing to all four central tasks of the H&T mission.

Our research perfectly aligns with the H&T themes “*Human-centered AI for health*” and “*Technology-supported transition in healthcare*”, by developing (WP1 and WP2) and improving end-user adoption (WP3) of intelligent, individually tailored VR-based technologies across the **entire health journey** and by assisting the transformation of the mental healthcare system into a more data-driven, resilient, inclusive, and affordable system. By educating students and junior researchers within the ecosystem of our expertise center, we also ensure that the next generation of professionals are equipped with the

interdisciplinary competencies and skills necessary to create the future of the *Me-Health InnoVatoRs* for VR-based applications across mental healthcare.

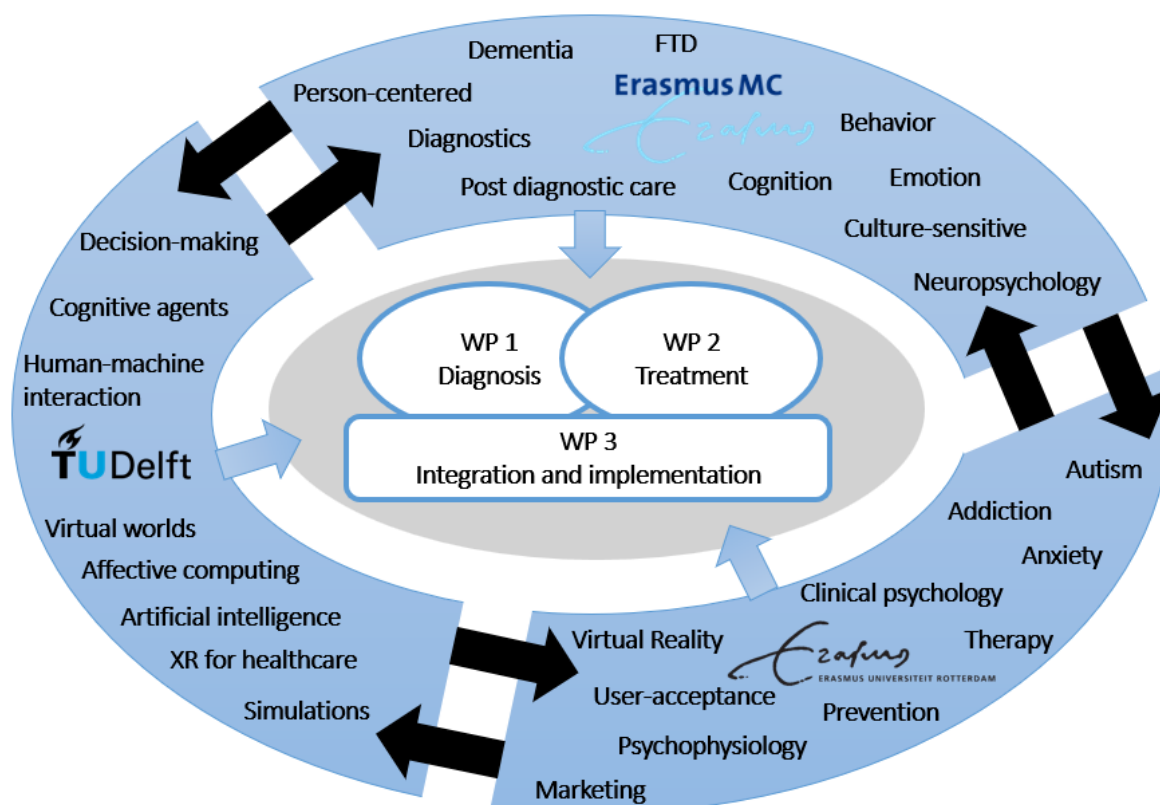
2.b Synergy between disciplines 392 words

Our newly established multidisciplinary consortium is the best fit for an effective realization of the development and use of VR solutions in mental healthcare. We will converge our expertise in artificial intelligence (AI), computer graphics and visualization, data science, clinical psychology, neurology, neuropsychology, and marketing, to establish intelligent and individualized VR-based diagnosis and treatment approaches that can ultimately be transferred to other mental disorders, and user-friendly and accessible at-home solutions (Figure 1).

The **Clinical Psychology group** and the **Alzheimer Center (AC) Erasmus MC** will provide expertise in detailed analysis of behavioral, neuropsychological, and psychophysiological patterns in mental disorders. Moreover, the access to patient groups and clinical experts via members of the psychology and neurology teams will integrate the very important perspective of end-users (patients and clinicians) early-on and thus incorporate co-creation to ensure high user-friendliness and technology adoption. The clinical input will allow the research groups of **TU Delft** to develop the VR-based approaches through data integration and real-time adaptation methods for VR environments and virtual agents, according to the patients' emotional state and behavior, as well as the context of the mental healthcare scenario.

With the objective of providing human-centered advanced and adaptive VR solutions for mental healthcare, the team of **Decision Making** contributes to *Me-Health InnoVatoRs* via their expertise in human-machine interaction, human-centered AI, and autonomous decision making of cognitive agents based on mathematical cognitive models of humans that estimate and predict their unobservable state-of-mind, and by adding real-time computationally efficient sensor fusion and soft sensor methods. In addition, the **Multi-Actor Systems** group brings in expertise in affective computing, including sensing and automatic recognition of affective states based on observable behaviors and bio-signals, which support diagnosis and the personalization of treatments. The **Computer Graphics and Visualization (CGV) group** brings strong expertise in generation of virtual worlds, efficient display of virtual content as well as perceptual rendering, and visual analysis of data as well as XR applications for healthcare. Complementary to this, **EUR Marketing** contributes with expertise on marketing strategies of new high-tech products, and contributes to the goal of end-user adoption and diffusion of new technologies, that result in communication strategies that lead to a fast new product takeoff.

Altogether, medical, psychological, technical, and marketing sciences will converge in this flagship to promote a game-change towards effective, science-based, personalized, and cost-efficient mental healthcare for all.

Figure 1. Synergy between disciplines.


2.c Long term sustainability plan 529 words

5-YEAR FEASIBILITY

Commitment from consortium members

The feasibility of developing an expertise center for VR in mental healthcare is driven by the commitment of our multidisciplinary team and established partnerships with industry, patient organizations, and healthcare institutions. While the proposed research is very ambitious, it is carefully built upon a strong clinical infrastructure combined with scientific expertise in various technical aspects of VR, mental health disorders, and marketing and innovation. All three institutions involved are highly committed to *Me-Health InnoVatoRs* flagship:

- EUR ESSB/RSM has invested 57k in VR facilities in 2022 and intends to commit another substantial amount (2023: approx. 13k).
- The AC of the Erasmus MC has made innovative diagnostic technologies as one of their long-term goals and provides additional commitment in terms of half a PhD and postdoc position (about 300k €).
- The Computer Graphics and Visualization (CGV) group at TU Delft provides additional commitment in terms of one PhD position (about 250k €) as well as 10k € for equipment.

Embedding in ongoing research initiatives

Our flagship is embedded and profits from recent initiatives at the EUR, such as the Erasmus Center for Data Analytics (ECDA) with expert groups on Virtual Reality and Artificial Intelligence and Data Science, and at the Erasmus MC, with the set-up of the Smart Health Tech Center (SHTC) with an expert group on mental healthcare. We will make use

of these facilities and their communication channels to raise awareness for our project within a broad group of researchers, mental healthcare organizations, and potential industry partners (see Table 4). Furthermore, our two showcase projects (for SAD and FTD) are embedded within research lines that have been running for over a decade at EUR and EMC⁵⁻⁸.

LONG-TERM SUSTAINABILITY

Long-term project plan

In the upcoming years, we will create an infrastructure for sustainable collaborations on the topic of intelligent VR-based solutions for mental disorders. Therefore, we will develop a sustainable Governance structure (see Figure 6), as well as devoting specific attention to the next steps in the 'value chain' (see Figure 2) throughout the project. The long-term sustainability will be a central theme for the executive board and in WP3, for which we will develop **a sustainability and mitigation plan** considering long-term funding, personnel, and translational research.

Translational research

The prototypes and knowledge created in the first five years will provide the necessary foundation for extending our solutions to other mental disorders (addiction, PTSD, autism), and at home solutions. To consider this future perspective, we have involved relevant researchers and clinicians throughout our project and in our end-user board (co-creation) and will make this a recurring theme in our consortium meetings.

Financing beyond the 5-year funded period

Me-Health InnoVatoRs involves several industry partners (see Figure 5), who are potential sponsors of VR research in the years beyond the first 5 years. Moreover, we will consider **long-term funding** by applying for new collaborative research grants (e.g., NWA-ORC, EU Horizon), intramural research grants (e.g. Vital Cities, Vital Citizens initiative at the EUR), as well as personal grants (i.e., 1 NWO VENI, 3 VIDI, 1 NWO Open competition and 1 ERC Starting grant), summing up to a potential amount of approximately 12 million Euro.

2.d Visibility, attracting talents and new partners word count: 449 words

A communication and dissemination board and education board will be installed that are responsible for increasing the visibility of our consortium and attracting new talents and partners (see Table 1).

Table 1. Measures to increase visibility and attract talents and partners

Target group	Objectives and tools/channels
Students	1) VR technologies for mental healthcare will be incorporated in the curriculum of existing BSc. and MSc. programs by, e.g., organizing guest lectures from consortium partners. Examples are: Psychology: - BSc: Anxiety (EUR)

	- MSc: Digital Interventions (EUR) Medicine: - BSc: Dementia (EMC) Technology: - MSc: Optimization for Systems and Control (TUD) - MSc: Mathematical and Human-Inspired Decision Making (TUD) - BSc: Clinical Technology (TUD&EMC) Management: - MSc: Marketing of New Technologies (EUR) 2) PhD course: we will set-up a course on transdisciplinary VR applications for mental health (via Erasmus MC Graduate School). 3) Summer school for junior researchers in the flagship: multidisciplinary lectures and workshops by experts and industry partners. 4) Collaborative thesis projects (BSc., MSc., PhD), including lab rotation to inspire students from different disciplines to interact with and learn from each other. Together, these initiatives will be distributed via the online <i>Me-Health innoVatoRs</i> platform and stimulate early interest in the field and strengthen an inflow of talent through PhD programs.
Scientific organizations & research groups	1) Top-level publications 2) Participation in international symposia/conferences, e.g.: - Psychology: EACLIPT, APS, STAR Society, NIP - Neurology: AAIC, ICFTD, FESN, Alzheimer Europe, GENFI consortium meetings, NVN - Technology: IEEE Virtual Reality, SIGGRAPH, ACM SIGGRAPH Symposium on Interactive 3D Graphics and Games, AAAI Conference on AI, IEEE CASE, Benelux Meeting on Systems and Control 3) A bi-annual meeting with the scientific advisory board, including independent (inter)national scientific experts 4) dissemination of results through <i>Me-Health innoVatoRs</i> platform
Industry	1) Organization of a twice-yearly VR Mental Healthcare symposium 2) Making a documentary about the project's progress (illustrating the healthcare problems, development of the VR solutions, and patient and clinician perspectives) 3) Communicating progress through social media platform for <i>Me-Health InnoVatoRs</i>, where results and informative videos will be shared 4) Enabling and facilitating public-private, cross-industry and cross-disciplinary knowledge flow via the SHTC Erasmus MC and ECDA EUR

	5) Giving presentations at industry-science symposia (e.g. technology-summit)
Societal partners/patient advocacy groups/end-users: mental health professionals/patients	1) Documentary (See “Industry”) 2) <i>Me-Health InnoVatoRs</i> social media outlets and platform 3) Social media channels, newsletters and websites of: • Alzheimer Nederland • FTD lotgenoten • De Nederlandse Vereniging voor Autisme (NVA) • PsyQ • Erasmus MC (AC Erasmus MC, Amazing Erasmus) • TU Delft (New Media Centre) • EUR (ESSB, RSM) 4) Bi-annual meeting with an end-user advisory board (see Figure 6) 5) (bi)annual newsletters from the AC Erasmus MC and FTD-RisC project

2.e Long term societal impact word count: 716 words

IMPACT

Healthcare impact

Mental disorders have devastating social and economic effects on individuals, families, and communities with one in every two people experiencing mental illness in their lifetime¹. As such, the total burden of mental disorders is enormous. Besides the impact on individuals' quality of life, a report from the OECD estimates the total costs of mental ill-health at over €600B billion across EU28 (27+UK) countries². In-patient treatment services are one of the largest drivers of this cost, resulting from slow diagnoses, poor treatment options, and low treatment accessibility². Through the *Me-Health InnoVatoRs* consortium, we will deliver VR-based solutions to enable more timely diagnosis and effective and remote treatment for mental disorders, and will eventually extend our developments for at-home solutions. These will significantly reduce the burden for the healthcare system and core costs of mental illness, caused by staff shortage and long waiting lists (see Figure 2).

Scientific impact

Successful implementation of the *Me-Health InnoVatoRs* research will provide a critical proof-of-principle with widespread potential impacts on a multitude of mental health disorders. From a scientific perspective, we expect to dramatically enhance our knowledge on the 1) behavioral/psychological characteristics of patients with FTD and SAD, 2) use of intelligent VR-based diagnostic and treatment approaches, and 3) factors that stimulate/hinder the acceptance of VR for diagnosis and treatment (see Figure 2). VR offers high ecological validity, which makes our results more reliable and useful for real-life and at-home applications^{3,4}. The knowledge that will be developed will provide the scientific groundwork for personalized diagnosis and treatment for anxiety and dementia, based on more accurate assessments of the underlying mechanisms.

Societal impact

Me-Health InnoVatoRs will provide user-friendly and cost-effective solutions for patients and professionals. In the long term, this will also result in highly personalized, accessible, and cost-effective prevention of chronic mental health problems. The ultimate societal impact will be a decrease in patients entering specialized mental healthcare, shorter waiting lists, and thus a significant reduction in individual and societal costs of mental health problems. The impact of the flagship on society is significant given it will enable earlier diagnosis, increased treatment options, improved treatment accessibility and higher treatment success rates (see Figure 2).

Technology Readiness Levels

VR-based diagnostics and treatment for mental healthcare, in general, is currently at technology readiness level **(TRL)4**. Although there are established companies that have started to work on, e.g., VR for reducing anxiety prior to surgeries⁹, or VR for testing navigational skills in dementia patients¹⁰⁻¹², these solutions are for short-term elimination/characterization of specific symptoms and do not consider systematic approaches for long-term personalized diagnosis and treatment approaches for anxiety and dementia. With *Me-Health InnoVatoRs*, we plan to reach **TRL6** by the end of the 5th year of the project, meaning that we will develop and prove the feasibility of our proposed solutions. Our long-term plans (~10 years) are to reach **TRL8**; with the support of our industrial partners, we will make the VR solutions for diagnosis and treatment of anxiety and dementia complete and operational.

SOCIETAL/ECONOMIC VALORIZATION STRATEGY

Realizing the expected societal and economic impact relies on successful market access. In turn, this is dependent on product development, regulatory approval, recognition from the medical community, reimbursement from health insurances, and engagement of distribution partners. Thus, the consortium's pathway to impact involves engaging industrial partners, key opinion leaders, insurers, and governmental healthcare departments.

Regulatory

To bring the VR solutions successfully to the clinical and foster valorization, all components will be validated in the laboratory. Industrial partners will be engaged to initiate product development, necessitating CE mark approval and adherence to the Medical Device Regulation of the EU. The consortium will engage clinical experts and advisors to design a suitable R&D framework in compliance with regulatory requirements.

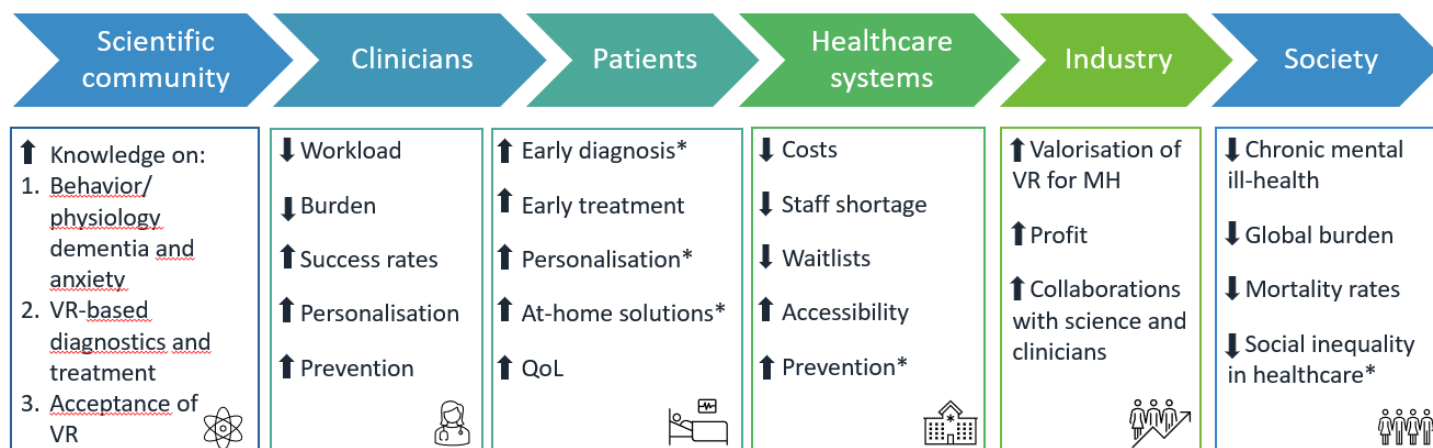
Reimbursement and entry to clinical guidelines

Negotiation of reimbursement is a key strategy for successful implementation of medical products. As such, the consortium will engage health insurers and governmental healthcare departments during the flagship to increase awareness and pave a route for 1) the implementation of VR within clinical guidelines, and 2) insurance reimbursement.

Broad stakeholder acceptance

Broad stakeholder acceptance will be achieved via the strategy laid out in section 2d and WP3. This will importantly include targeting of industrial partners, patient organizations, and end-users (mental health-care providers and patients).

Figure 2. Value chain to societal impact



*refers to themes that align with the H&T mission

3. Flagship proposal (25%) max 2000 words for 3.a to 3.d – word count: 1993

3.a Ambitions word count: 266 words

We will establish a **national expertise center** on the topic of intelligent VR in mental healthcare, that is based on a **sustainable collaboration** between knowledge institutes, industry, healthcare providers, patients, and clinicians with the main ambition of performing excellent Convergence Health & Technology research to:

Improve early diagnosis and treatment across mental disorders by developing and implementing personalized intelligent VR-based solutions

The main goal of *Me-Health InnoVatoRs* is to change the current mental health diagnostic and treatment landscape to **improve quality of life** for patients with mental disorders. Current diagnostic and treatment options are based on standard paper-and-pencil, and subjective evaluations, and fail to encapsulate the richness and complexity of real-life situations. Furthermore, diagnostics and treatment are based on a “one-size-fits-all” principle rather than a personalized approach. While VR is potentially able to provide solutions to tackle these challenges^{3,4}, there is a significant gap between VR solutions and the clinical feasibility and adoption by end-users (patients/clinicians). *Me-Health InnoVatoRs* contributes to these challenges, and thereby perfectly fits the H&T mission of facilitating new opportunities for **prevention, diagnostics, cure, and care** in mental disorders. Our consortium will focus on dementia and anxiety, two of the most common and costly mental disorders^{13,14}.

Our ambitions are **innovative and challenging** since they promote a transition from conventional mental healthcare systems into human-centered intelligent systems.

Challenges lie in the convergence of technological developments and personalized approaches in vulnerable patient populations. For this, we will rely upon an integrated and co-creation-based approach involving various end-users in each development step. This ultimately leads to extending VR-based solutions to other mental disorders, and at-home solutions.

3.b Aims and feasibility word count 770 words

Aims

Me-Health InnoVatoRs will establish an expertise center where health and technology converge in order to:

- 1) perform excellent research for improving early diagnosis and treatment across mental disorders using intelligent VR solutions,
- 2) guarantee acceptance among the industrial sector and end-users by involving stakeholders and end-users via co-creative approaches,
- 3) promote long-term sustainability of *Me-Health InnoVatoRs* by involving the next generation of clinical and scientific professionals necessary to create the future of VR applications across mental healthcare.

Research activities

Short term (5-year) aims include:

WP1: Developing VR diagnostic tools for socio-affective issues in FTD and SAD

WP2: Developing VR treatment tools for personalized and adaptive stress reduction in FTD and SAD

WP3: Enhancing end-users' adoption and implementation of VR diagnostic and treatment methods

Long-term (10-year) aims include extending VR diagnostic and treatment approaches to **other mental disorders** (e.g., addiction, PTSD, autism), and developing **at-home treatment solutions**. To guarantee the achievement of these long-term goals, we use integrative approaches that involve end-users and stakeholders, and collaborate with experts in human-centered social AI (TUD/TNO).

Involvement of students and early career researchers

An **Education board** will be installed to ensure involvement of students and early career researchers in the expertise center, as well as their personal and scientific development. Our PhD students and postdoctoral researchers will collaborate closely across universities, performing the research projects, and assist with the scientific management of *Me-Health InnoVatoRs*, as three Bachelor and Master students from Psychology, Medicine, Marketing, Clinical Technology, and Systems and Control programs will be involved by conducting relevant **BSc and MSc thesis projects** under the supervision of

our interdisciplinary team, and by **lab rotations** across EMC, TUD, and EUR (see Table 1 for details).

Feasibility

The research performed within our Flagship is highly feasible as it is built upon **extensive clinical and technological expertise** (i.e., AC clinic, and Erasmus Behavioral Lab), and a clinical, research, and marketing **infrastructure** (i.e., AC/GENFI, VR lab, ECDA). The proposed aims are ambitious, but with careful consideration of the team composition, including the intended staff (i.e., PhD students, postdoctoral researchers, and a project manager), and a firm governance structure, both short and long-term objectives are feasible. The project manager will be responsible for our governance and overall infrastructure (see Figure 6) and will be in direct contact with our steering group (overall management) and postdocs (scientific management).

Strategy to attract external funding

We have concrete plans for attracting external funding, by grant applications at ERC and NWO (individual grants), and a larger NWA-ORC grant at the end of the 5 years (see 2c). Furthermore, extending collaborations with current strategic business partners (see section of partners), and attracting new (business) partners is an important part of our **long-term sustainability and mitigation plan**.

Risks

We have identified risks and developed mitigation strategies, subdivided into:

Technological risks: 1. Hardware, Software and Data Security risks, 2. Uncertainties in interactions of patients with VR, 3. Unrealistic Virtual environments. These risks will be mitigated by making use of the expertise of the TU Delft Decision Making group and CGV group in incorporating and efficiently handling sources of uncertainties in modeling and efficient data visualization. In addition, the expertise of medical partners will be brought in, and we will use co-creation strategies and patient-centered approaches, ultimately incorporating the feedback of end-users. Risks considering data security will be mitigated by involving data security specialists early on in the project.

Operational risks: 1. The inclusion of a diverse, and representative patient population, for which we will collaborate with the outpatient cross-cultural memory clinic (AC Erasmus MC, PI Dr. Papma¹⁵). In case of FTD, we will make use of a longitudinal cohort study at the AC Erasmus MC and will include participants that have already agreed upon yearly research visits (FTD-RisC, as part of the Genetic FTD Initiative (GENFI)^{5,6}). Existing contacts with mental healthcare providers (PsyQ) will allow for prototype pilot validation in patients with SAD. We will account for fluctuation of research personnel, by establishing careful documentation on ways of working, codes, quality assurance, etc.

Regulatory risks: 1. Losing cohesion within WPs and subtasks. To increase cohesion, we will not only rely on our Governance structure but have positioned a postdoc at each university that will be appointed scientific coordinator and will function as the central point within the convergence. 2. Consortium partner performs below expectation, or in conflict. We will overcome this by regular communication and solution seeking. If necessary, we will fall back on the conflict resolution procedure as described in the consortium agreement.

Financial risks are not foreseen in the upcoming five years, but long-term financial risks are mitigated by a long-term sustainability plan.

3.c Work plan word count: 855 words

Figures 5 and 6 illustrate Gantt charts with the 10-year research plan and 5-year consortium activities.

Table 3. Description of work packages

WP1 VR diagnostics for mental health		Duration 2022 - 2032	Main leads: EMC: Dr. Seelaar EUR: Dr. Andreatta TUD: Prof. De Schutter
Five-year plan: FTD & SAD	In the first five years we will develop VR-based diagnostic tests for analyzing socio-affective processes in FTD and SAD. FTD is an early onset type of dementia with an atypical onset with behavioral changes and social cognition deficits, which are hard to capture in a clinical interview. Similarly, SAD is difficult to diagnose when relying only on self-reports and excluding observations of actual social behavior (e.g., gaze avoidance).		
Ten-year plan: other mental disorders	After five years , the virtual environments, visualization modes, cognitive models, decision making algorithms, and data analysis approaches will be extended for diagnostic purposes in other mental disorders (e.g. addiction, PTSD, autism).		
Deliverables	VR-based diagnostic systems that are personalized, adaptive, human-centered , and capable of autonomously functioning in line with diagnostic goals of FTD and SAD.		
Activities	1. Developing cognitive models of patients with FTD and SAD 2. Developing intelligent virtual environments and their efficient VR-based visualization for diagnostic interventions 3. Pilot projects on standardized social cognition and behavior diagnostics in students for FTD and SAD.		
WP2: Personalized VR therapy for mental health		Duration: 2022 - 2032	Main leads EMC: Dr. Papma EUR: Prof. Wieser TUD: Prof. Eisemann
Five-year plan: FTD & SAD	In the first five years we will develop individualized VR-based treatment and training approaches to reduce stress in SAD and FTD patients. For this we will apply live adaptation of VR environments according to the behavioral and physiological responses (e.g., heart rate) responses of patients. Recent developments using VR therapies have shown that VR is feasible in various therapeutic fields (e.g., exposure therapy). However, personalization and adaptability of these approaches are still in their infancy.		
Ten-year plan: other mental disorders	After five years special attention will be given to the next steps of the value chain (see Figure 2) by focusing on: 1) extending the therapeutic VR-based approaches to other mental disorders, such as addiction, PTSD and autism 2) developing at-home solutions for diminishing long waiting lists for low-level treatment options and for giving access to hard-to-reach populations.		

Deliverables	VR-based treatment systems that are personalized, adaptive, human-centered , and capable of autonomously functioning in line with treatment goals.	
Activities	1. Developing computationally VR-based behavior support agents and environments, harmonized to the human therapist decision-making process, and adaptive to patient and context 2. Building highly adaptive VR environments for participants with SAD and genetic FTD patients.	
WP3: End-user adoption and sustainable impact	Duration: 2022 - 2032	Main leads EMC: Dr. Poos EUR: Prof. van Everdingen TUD: Prof. Neerincx
Five & ten year plan	Stakeholders (professionals, caregivers, industry) and end-users (adolescents, patients, and clinicians) will be involved in all steps of the developmental, scientific, and impact processes via co-creation . We will integrate the insights from WP1 and WP2 to improve the acceptance and engagement of end-users and will develop implementation and generalization strategies that provide a fundamental foundation for long-term sustainability of the project. Factors that limit adoption of current VR-based treatment approaches will be identified and provided with effective solutions that increase high end-user satisfaction and adoption of these technologies. As an integration and sustainability hub, WP3 will build and maintain the necessary technological infrastructure and collaborative frameworks for sustainable impact , including interdisciplinary expertise .	
Deliverables	1. Framework on how to improve acceptance and engagement of end-users of VR diagnostic and treatment applications 2. Implementation/generalization strategies for VR diagnostic and treatment applications across mental disorders	
Activities	1. Examining, identifying, and classifying the characteristics and needs of end-users, as input for WP1 and WP2 2. Investigating factors that stimulate/hinder the acceptance of VR-based diagnosis and treatment from the perspective of clinicians and patients 3. Extracting generalized knowledge from the showcase studies of WP1 and WP2 to develop generalized cognitive models, decision making approaches, design patterns for virtual agent's coaching, adaptivity and personalization algorithms, and to prepare fundamental basis for long-term plans of <i>Me-Health InnoVatoRs</i>.	

Data and infrastructure plan

We will act in compliance with the General Data Protection Regulation (GDPR, 2018), the Netherlands Code of Conduct for Research Integrity (2018), and the policies of the involved institutes considering data management. Pseudonymised data will be safely stored on the Research Storage and Compute Services (EMC), and the EUR Document Vault, both end-to-date end-encrypted and ISO 27001 certified secure environments. Access will be limited to research personnel involved. Data sharing between the three

institutes will be done via a XNAT server, and the process will be described in the consortium agreement. Data will be archived using an *Interoperable* format to allow for reuse. We will share the metadata in an online open access repository, making it *Findable*. Based on the metadata a data request can be done. After approval by the principal investigator, researchers can download the data (*Accessible*).

Required facilities

VR lab EUR, TU Delft computational facilities and initial cognitive models and data analysis algorithms available for developing cognitive agents and environments, Erasmus MC cohort study.

3.d Alignment of the program with relevant national and international research agendas word count: 102 words

This program aligns well with several national (NWA, AI Alliance) and EU research agendas (JPND) on significantly enhancing the use of AI-based methods in various applications, as well as with HITOP, RdOC initiatives for personalizing treatment of mental disorders transdiagnostically. Furthermore, the Knowledge and Innovation agenda (KIA 2020-2023) from NWO and the ministry of Economic Affairs and Climate Policy encourages public-private consortia to contribute to Dutch innovation policy, e.g. using artificial intelligence for major societal problems (e.g. dementia). Furthermore, the National Dementia Strategy (2021-2030), by the ministry of Health, Welfare and Sports, acknowledges both technological innovation and young onset dementia (e.g. FTD).

4. Consortium (25%) max 2000 words for 4.a to 4.d

4.a Track record of the consortium members word count: 440

The Alzheimer Center (AC) Erasmus MC is an expert center for dementia diagnostics and research into its causes, treatment, and post-diagnostic care, and officially declared an expertise center on FTD by the Dutch Ministry of Health. They have a strong foothold regarding translational research with great societal impact, focusing on diagnostic challenges in underserved populations, such as rare dementias like FTD, and non-Western immigrants. Their impact is reflected by several personal awards (Young Outstanding Researcher award from Alzheimer Nederland, ZonMw-Alzheimer Nederland research fellowship), and the fact that members are PI of large (inter)national projects (ZonMw, JPND, Bluefield Project to Cure FTD, GENFI), and involved in national private-public research consortia (ABOARD consortium, Nationale Dementie Strategie). They train (neuro)psychologists and clinicians in dementia diagnostics, care, and research (BSc, MSc, PhD, specialist training psychologists, neurologists), for which several prizes were awarded (Deelman prize, Psychology Teacher of the Year award). **The Clinical Psychology group** from EUR aims to unravel the mechanisms underlying the development of, maintenance of, and recovery from mental disorders (e.g., anxiety, addiction, autism, etc.). Their expertise is interdisciplinary, from neuroscience to clinical practice, and is strongly involved in educating the next generation of researchers and clinicians. The group has a significant track record in attracting external funding (ERC,

NWO, FWO, DFG, NIDA/NIH). Members are represented in relevant journal boards and elected board members of international scientific societies (SPR, ESCAN). They have received prestigious awards for research (SPR Early career award, Young academy Europe), and teaching (Best PhD supervisor). The **Decision Making group**, **CGV group** and **Multi-Actor Systems** group focus on various fundamental aspects of 3D modeling, data visualization, human-machine interaction, autonomous adaptive decision making approaches, conversational agents, virtual cognition, socio-cognitive engineering, processing, interpretation and securement of data, and affective computing. They actively seek research and technology-transfer collaborations with companies and societal organizations, and applications varying from healthcare to automated systems. The impact of these groups is reflected by several external grants from international and national funding agencies (ERC Advanced, multiple EU-funded projects, NWO VIDI and Veni), prestigious awards (2018 IEEE ITSS Best PhD Dissertation Award, IEEE Fellow, Eurographics Young Researcher Award and the Netherlands Prize for ICT Research 2019), (senior) editorships of high-impact journals (IEEE Transactions on ITS, IEEE Transactions on Automatic Control, Automatica), as well as the organization of major conferences (e.g. IEEE ITSC, IFAC ADHS, IFAC CTS, Eurographics, EGSR). Furthermore, members from these groups have been involved in various committees and editorial boards of international scientific societies (IFAC, IEEE). All consortium members publish in the most prestigious journals in their fields, are cited widely (>70.000 citations), and have an excellent track record in supervising master, bachelor, and PhD students.

4.b Expertise of the consortium members in relation to the aims and objectives of the program word count: 525

The expertise of all consortium members provides a perfect fit to the aims and objectives within this flagship (see references). Expertise from the **AC Erasmus MC**¹⁵⁻²¹ on the determinants of cognition, emotion and behavior in patients with dementia, most specifically FTD (Dr. Esther van den Berg, Dr. Harro Seelaar, Dr. Jackie Poos, see references), and person-centered dementia diagnostics and care (Dr. Janne Papma, Dr. Esther van den Berg) will be invaluable for the researchers from TUD to develop VR scenarios for improving diagnostics and care in FTD. These deliverables will then be tested and eventually implemented by experienced clinicians (Dr. Harro Seelaar, Dr. Esther van den Berg) that are involved in the project from the beginning. Expertise from the **Clinical Psychology group**^{7,8,22-25} on VR research to investigate the cause and symptoms of anxiety disorders (Dr. Andreatta, Prof. Wieser), experimental testing of effects of interventions on cognition and behavior (Dr. Cousijn, Dr. Van der Hallen, Prof. Wieser), and the psychophysiological and behavioral correlates of mental disorders (Dr. Andreatta, Dr. Cousijn, Dr. Van der Hallen, Prof. Wieser) will contribute to develop and test diagnostic and treatment tools for respective disorders. Clinical experience (Dr. Van der Hallen) will help in co-creating VR intervention tools which will be adopted by clinicians. The latter will also be greatly improved by the research expertise of the marketing group (Prof. van Everdingen) on implementing new technologies and conducting end-user satisfaction research. The **Decision Making group**²⁶⁻³¹ (Prof. dr. De Schutter; dr. Jamshidnejad, Prof. dr. Neerincx) has extensive expertise in autonomous decision making of machines, human-machine interaction, mathematical models of

human's cognition, real-time intelligent data fusion and soft sensing, human-centered co-design of social AI and personal e-coach, e.g., interactive social agents for dementia patients and cognitive agents for autism therapeutic intervention. The expertise of the **CGV group**³²⁻²⁶ (Prof. dr. Elmar Eisemann, Dr. ir. Rafael Bidarra, Dr. Michael Weinmann, Dr. Ricardo Marroquim) regarding the generation of virtual worlds, the efficient display of virtual content, and the visual analysis of data overlaps with many current trending fields like data science, AI, AR/VR, 3D computer vision, computational photography, high-performance and GPU programming. In the scope of this project, this expertise as well as the expertise regarding the development of serious games for societal topics will provide the foundation for the development of intelligent virtual environments, their corresponding rendering as well as their quick adaptability to the individual needs in the current therapeutic treatment. The **Multi-Actor Systems group**³⁷⁻³⁹ brings in strong expertise on affective computing (Dr. ir. Iulia Lefter), which is highly relevant for the awareness of patients' affective states. This benefits diagnostics by providing objective behavior quantification, and personalized therapy by giving inputs to the highly adaptive virtual environment.

4.c Involvement of partners word count: 217

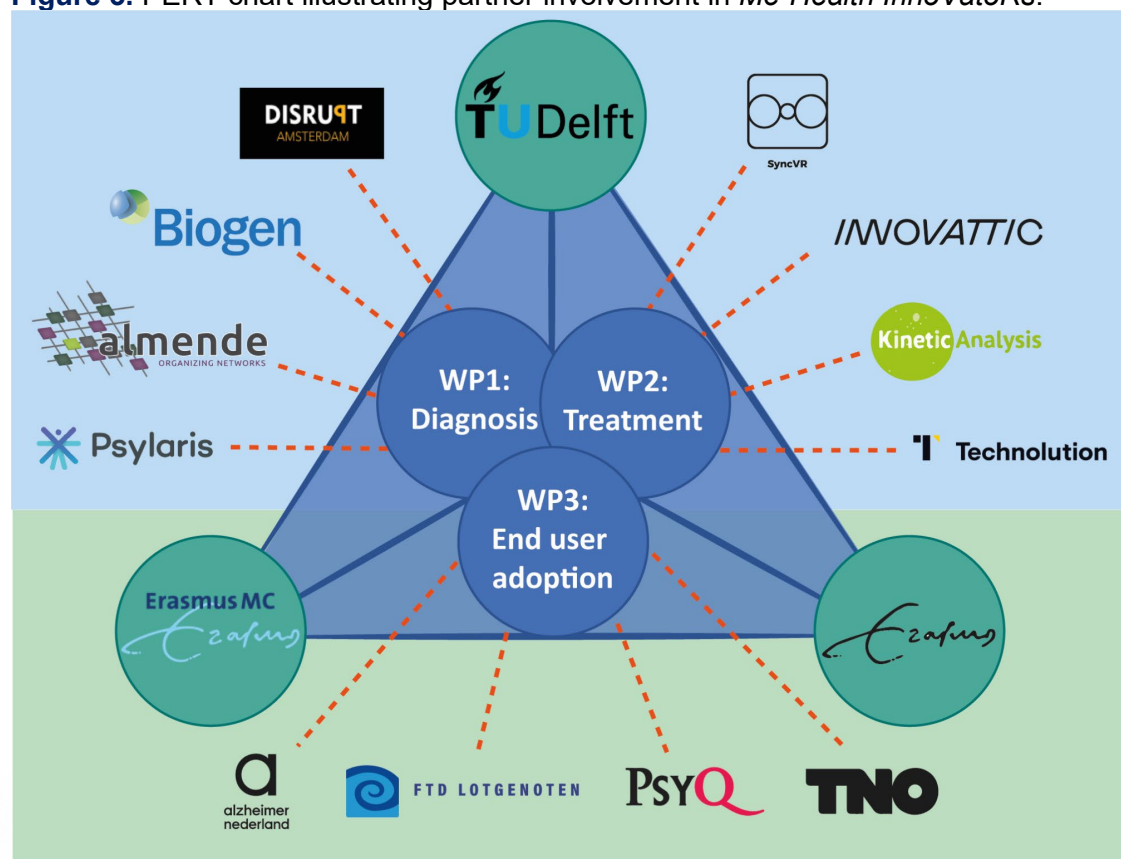
Our public-private consortium of scientific, societal, and industry partners will provide the best fit for realizing an expertise center on intelligent VR solutions in mental healthcare. We involve external partners that are highly committed to the flagship, and involved in all different stages of the consortium (see letters of intent). This is demonstrated in Table 4 and Figure 5.

Table 4. Involvement of partners in the ecosystem of *Me-Health InnoVatoRs*

Partner name	WPs	Contribution	Contact person
<i>Private and semi-private</i>			
Biogen – digital health solutions	1, 2	End-user advisory board	Georgia Mitsi, Alexandra Auffret
Almende	1, 2	€10.000 in-kind End-user advisory board	Aad Nales
Technolution	1, 2	€60.000 in-kind End-user advisory board	Serge de Vos
Disrupt	1, 2	€15.000 in-kind	Dimitra Kontogiorgou
Kinetic Analysis B.V.	2	€10.000 in-kind (WP2)	Maarten Gijssel
InnoVattiC	1, 2	€6000 in-kind	Lauwerens Metz
SyncVR Medical	1, 2	€50.000 in-kind	Robert Brouwer
Psylaris	1, 2	€40.000 in-kind	Mike Verhiel

Public partners			
PsyQ	1-3	€75.850 in-kind End-user advisory board	Annet Spijker, prof. dr. van der Heiden
Alzheimer Nederland	3	€10.000 in-cash €2000 in-kind	Marco Blom, Dinant Bekkenkamp
FTD Lotgenoten	3	End-user advisory board	Henk Jan van der Waal, Saskia Daanen
Scientific partners			
Genetic FTD Initiative (GENFI)	3	Advisory council (scientific)	Prof. dr. Rohrer
Nederlandse Organisatie voor Toegepast Natuurwetenschappelijk onderzoek (TNO)	1-3	Advisory council (scientific)	Prof. dr. Neerincx

Figure 3. PERT chart illustrating partner involvement in *Me-Health InnoVatoRs*.



4.d Governance, organization and collaboration strategy of the consortium word count 524

Me-Health InnoVatoRs is organized in such a way that it facilitates collaboration and makes optimal use of the expertise of all consortium members. To mitigate the risk of losing cohesion, we plan a management structure as depicted in Figure 6. This will facilitate active interdisciplinary collaboration and interaction among all consortium members, but at the same time also facilitates effective decision-making, progress monitoring and reporting. The bodies within the VR mental healthcare consortium are:

- **General assembly:** includes all consortium members and consists of all (or one representative) applicants and external partners. Prof. dr. Matthias Wieser will be the chair. The main decision items for this level are the composition of the consortium (e.g. replacement or addition of new partners), major project level changes (objectives, deliverables and budget changes), and major project risks that affect the course of the project. Meetings will be organized every 6 months (spring and autumn meetings).
- **Executive board:** includes one representative from each of the five research groups. Main tasks include monitoring the progress of all WP activities and deliverables, as well as budget, but also the coordination of the progress reporting and assessment at the WP and project level of each consortium partner. The steering group includes a junior researcher (post-doc) which rotates every year. The steering group will meet once a month.
- **Project management:** consists of three scientific coordinators (a post-doc from each institute) and a project manager. Tasks include the preparation and organization of consortium meetings, meetings with the end-user committee and the advisory board, as well as maintaining efficient internal project documentation, and social media outings. The project manager is also responsible for day-to-day management of administrative, contractual and financial issues, in close contact with project controllers at the three institutes, to monitor the project budget and financial reporting. The scientific coordinators will be the point of contact for all WP teams.
- **Scientific advisory board:** provides independent scientific expert advice in the form of feedback on results and deliverables of the consortium. Meetings will take place bi-annually, but specific members of this board are consulted depending on the issues at hand. The scientific advisory board includes prof. dr. Jonathan Rohrer (consultant neurologist at University College London, and PI of GENFI), prof. dr. Marjolein de Vugt (healthcare psychologist specialized in psychosocial technological innovations in young-onset dementias), prof. dr. Lucian Busoniu (expert on the field of robotics and multi-agent systems), prof. dr. Paul Pauli (expert in the field of anxiety disorders and virtual reality research).
- **End-user Committee:** will be highly involved during the project and provide input in a co-creational iterative process. The committee consists of the scientific coordinators, and a representative of patients and caregivers and professionals, as well as representatives from FTD Lotgenoten and PsyQ. Co-financing external partners (see Table 4 and Figure 5) can participate when applicable.

- **Education board:** will be responsible for all of the educational activities within the consortium (see Table 1). The committee consists of a representative from the three Institutes, and a rotating junior researcher.
- **Communication and dissemination board:** will be responsible for all communication and dissemination activities of the consortium (see Table 1).

Figure 4. Governance structure of *Me-Health innoVatoRs*

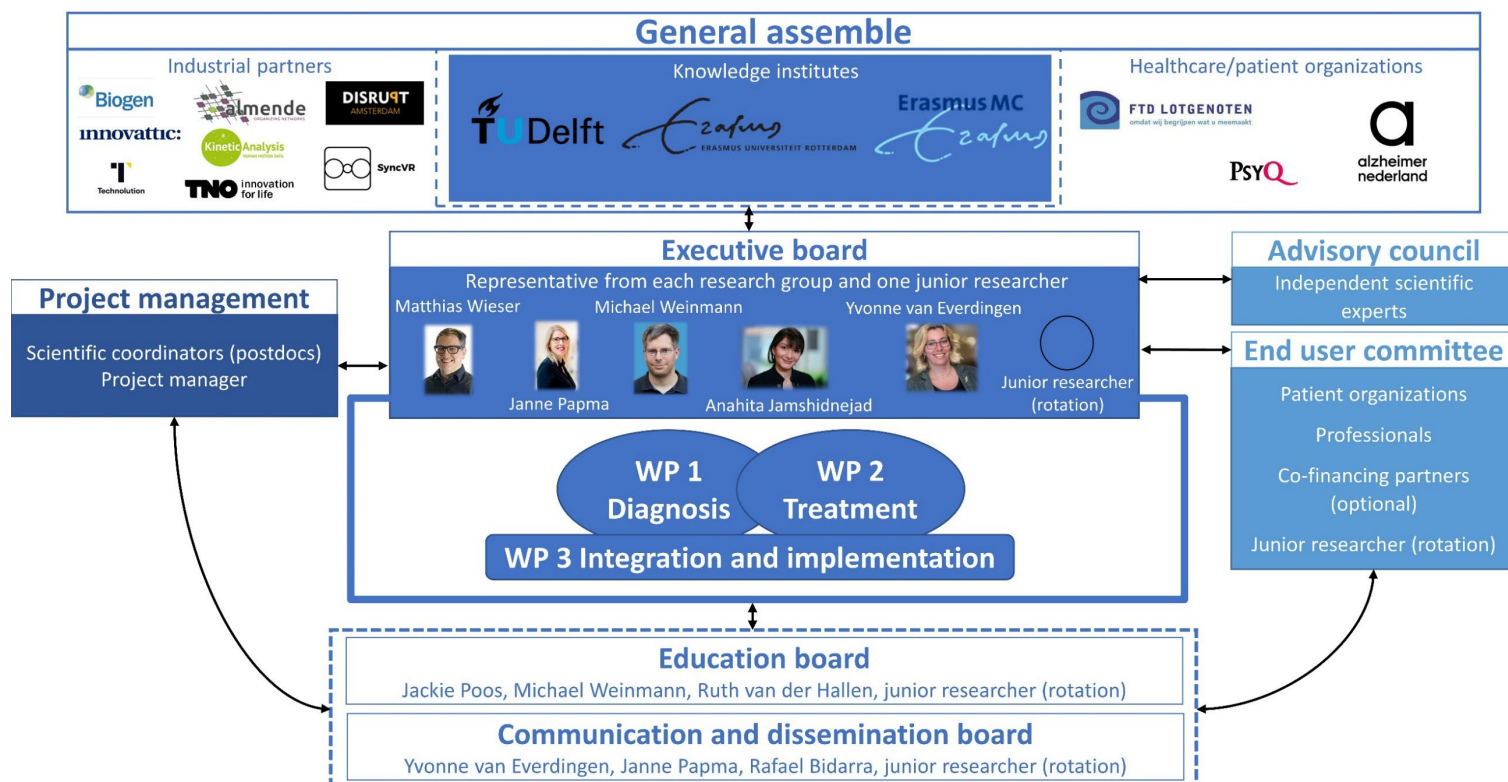


Figure 5. Gantt chart with the 10-year research plan

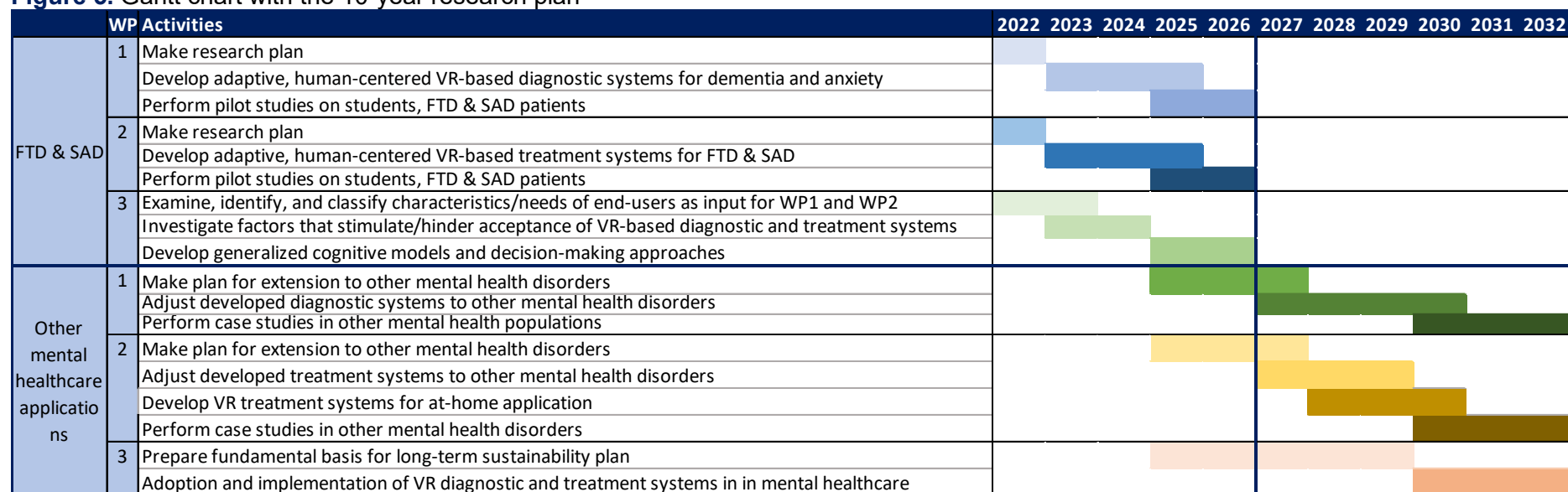
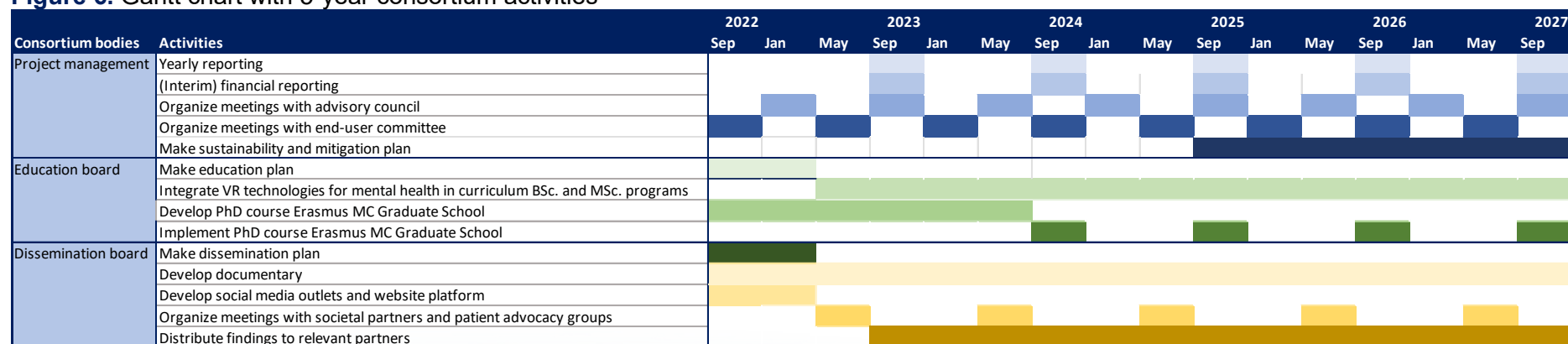


Figure 6. Gantt chart with 5-year consortium activities



Annexes (do not count toward maximum number of pages)

References

1. OECD. Mental Health. <https://www.oecd.org/els/health-systems/mental-health.htm>
2. OECD (22/11/2018). Mental health problems costing Europe heavily. <https://www.oecd.org/newsroom/mental-health-problems-costing-europe-heavily.htm#:~:text=Besides%20the%20impact%20on%20people's,across%20the%208%20EU%20countries.>
3. Freeman D, Reeve S, Robinson A, Ehlers A, Clark D, Spanlang B, Slater M. Virtual reality in the assessment, understanding, and treatment of mental health disorders. *Psychol Med*. 2017;47(14):2393-2400.
4. Emmelkamp PMG, Meyerbröcker K. Virtual Reality Therapy in Mental Health. *Annu Rev Clin Psychol*. 2021;17:495-519.
5. Rohrer JD, Nicholas JM, Cash DM, van Swieten J, Doppler E, Jiskoot L, et al. Presymptomatic cognitive and neuroanatomical changes in genetic frontotemporal dementia in the Genetic Frontotemporal dementia Initiative (GENFI) study: a cross-sectional analysis. *Lancet Neurol*. 2015;14(3):253-62.
6. Doppler EG, Rombouts SA, Jiskoot LC, Heijer Td, de Graaf JR, Koning Id, et al. Structural and functional brain connectivity in presymptomatic familial frontotemporal dementia. *Neurology*. 2013;80(9):814-23.
7. Andreatta, M., Neueder, D., Herzog, K., Genheimer, H., Schiele, M. A., Deckert, J., et al. Generalization of conditioned contextual anxiety and the modulatory effects of anxiety sensitivity. *Neurotherapeutics*. 2020; 17(3), 1239-1252.
8. Wieser MJ, Keil A. Attentional threat biases and their role in anxiety: A neurophysiological perspective. *Int J Psychophysiol*. 2020;153:148-158.
9. Eijlers R, Dierckx B, Staals LM, Berghmans JM, van der Schroeffer MP, Strabbing EM, et al. Virtual reality exposure before elective day care surgery to reduce anxiety and pain in children: A randomised controlled trial. *Eur J Anaesthesiol*. 2019;36(10):728-737.
10. Bayahya AY, Alhalabi W, AlAmri SH. Smart Health System to Detect Dementia Disorders Using Virtual Reality. *Healthcare (Basel)*. 2021; 28;9(7):810.
11. Poos JM, van der Ham IJM, Leeuwis AE, Pijnenburg YAL, van der Flier WM, Postma A. Short Digital Spatial Memory Test Detects Impairment in Alzheimer's Disease and Mild Cognitive Impairment. *Brain Sci*. 2021;11(10):1350.
12. Cushman, L.A.; Stein, K.; Duffy, C.J. Detecting navigational deficits in cognitive aging and Alzheimer disease using virtual reality. *Neurology* 2008;71, 888–895.
13. Aranda MP, Kremer IN, Hinton L, Zissimopoulos J, Whitmer RA, Hummel CH, Trejo L, Fabius C. Impact of dementia: Health disparities, population trends, care interventions, and economic costs. *J Am Geriatr Soc*. 2021;69(7):1774-1783.
14. Bandelow B, Michaelis S. Epidemiology of anxiety disorders in the 21st century. *Dialogues Clin Neurosci*. 2015;17(3):327-35.
15. Franzen S, van den Berg E, Bossenbroek W, Kranenburg J, Scheffers EA, van Hout M, et al. Neuropsychological assessment in the multicultural memory clinic: Development and feasibility of the TULIPA battery. *Clin Neuropsychol*. 2022:1-21.
16. Rohrer JD, Nicholas JM, Cash DM, van Swieten J, Doppler E, Jiskoot L, et al. Presymptomatic cognitive and neuroanatomical changes in genetic frontotemporal dementia in the Genetic Frontotemporal dementia Initiative (GENFI) study: a cross-sectional analysis. *Lancet Neurol*. 2015;14(3):253-62.
17. Doppler EG, Rombouts SA, Jiskoot LC, Heijer Td, de Graaf JR, Koning Id, et al. Structural and functional brain connectivity in presymptomatic familial frontotemporal dementia. *Neurology*. 2013;80(9):814-23.

18. Jiskoot, L. C., Panman, J. L., Meeter, L. H., Dopper, E. G., Donker Kaat, L., Franzen, S., et al (2019). Longitudinal multimodal MRI as prognostic and diagnostic biomarker in presymptomatic familial frontotemporal dementia. *Brain*, 142(1), 193-208.
19. Poos, J. M., Moore, K. M., Nicholas, J., Russell, L. L., Peakman, G., Convery, R. S., Cognitive composites for genetic frontotemporal dementia: GENFI-Cog. *Alzheimer's research & therapy*. 2021; 14(1), 1-12.
20. van den Berg E, Poos JM, Jiskoot LC, Montagne B, Kessels RPC, Franzen S, van Hemmen J, Eikelboom WS, Heijboer EGC, de Kriek J, van der Vlist A, de Jong FJ, van Swieten JC, Seelaar H, Papma JM. Impaired Knowledge of Social Norms in Dementia and Psychiatric Disorders: Validation of the Social Norms Questionnaire-Dutch Version (SNQ-NL). *Assessment*. 2021 15:10731911211008234.
21. Jiskoot, L. C., Poos, J. M., Vollebergh, M. E., Franzen, S., van Hemmen, J., Papma, J. et al. Emotion recognition of morphed facial expressions in presymptomatic and symptomatic frontotemporal dementia, and Alzheimer's dementia. *Journal of neurology*. 2020; 268(1), 102-113.
22. Stegmann Y, Ahrens L, Pauli P, Keil A, Wieser MJ. Social aversive generalization learning sharpens the tuning of visuocortical neurons to facial identity cues. *Elife*. 2020;9:e55204.
23. Andreatta M, Pauli P. Contextual modulation of conditioned responses in humans: A review on virtual reality studies. *Clin Psychol Rev*. 2021;90:102095.
24. Kroon E, Kuhns L, Hoch E, Cousijn J. Heavy cannabis use, dependence and the brain: a clinical perspective. *Addiction*. 2020;115(3):559-572.
25. Van der Hallen R, Jongerling J, Godor BP. Coping and resilience in adults: a cross-sectional network analysis. *Anxiety Stress Coping*. 2020;33(5):479-496.
26. T. Ascensão and A. Jamshidnejad. Autonomous Socially Assistive Drones Performing Personalized Dance Movement Therapy: An Adaptive Fuzzy-Logic-Based Control Approach for Interaction With Humans. *IEEE Access*. 2022;10:15746-15770.
27. L. Busoniu, R. Babuska, B. De Schutter, and D. Ernst. Reinforcement Learning and Dynamic Programming Using Function Approximators. Boca Raton, Florida: CRC Press. 2010.
28. R.R. Negenborn, B. De Schutter, and J. Hellendoorn. Multi-agent model predictive control for transportation networks: Serial versus parallel schemes. *Engineering Applications of Artificial Intelligence*. 2008; 21(3): 353-366.
29. K. Verbert, R. Babuska, and B. De Schutter. Combining knowledge and historical data for system-level fault diagnosis of HVAC systems. *Engineering Applications of Artificial Intelligence*. 2017; 59:260-273.
30. Tielman, M.L., Neerincx, M.A. and Brinkman, W.P. Design and evaluation of personalized motivational messages by a virtual agent that assists in post-traumatic stress disorder therapy. *Journal of Medical Internet Research*. 2021; 21(3):e9240.
31. Schoonderwoerd, T. A., Jorritsma, W., Neerincx, M. A., & Van Den Bosch, K. Human-centered XAI: Developing design patterns for explanations of clinical decision support systems. *International Journal of Human-Computer Studies*. 2021;154:102684.
32. M.L. Tielman, M.A. Neerincx, R. Bidarra, B. Kybartas, and W.P. Brinkman. A therapy system for post-traumatic stress disorder using a virtual agent and virtual storytelling to reconstruct traumatic memories. *Journal of medical systems*. 2017;41 (8).
33. R. Lopes, and R. Bidarra. Adaptivity challenges in games and simulations: a survey. *IEEE Transactions on Computational Intelligence and AI in Games*. 2011;3(2):85 - 99,

34. P. Stotko, S. Krumpen, M. Schwarz, C. Lenz, S. Behnke, R. Klein, M. Weinmann. A VR system for immersive teleoperation and live exploration with a mobile robot. *IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)*. 2019:3630 - 3637
35. P. Stotko, S. Krumpen, M.B. Hullin, M. Weinmann, and R. Klein. SLAMCast: Large-scale, real-time 3D reconstruction and streaming for immersive multi-client live telepresence. *IEEE Transactions on Visualization and Computer Graphics*. 2019;25(5):2102 – 2112.
36. P. Shi, M. Billeter, and E. Eisenmann. SalientGaze: Saliency-based gaze correction in virtual reality. *Computers & Graphics*. 2020;91:83-94.
37. Lefter, I., Burghouts, G. J., & Rothkrantz, L. J. Recognizing Stress using Semantics and Modulation of Speech and Gestures. *IEEE Transactions on Affective Computing*. 2015;7(2):162-175.
38. Lefter, I., Jonker, C. M., Tuentje, S. K., Veling, W., & Bogaerts, S. NAA: A Multimodal Database of Negative Affect and Aggression. In *2017 Seventh International Conference on Affective Computing and Intelligent Interaction (ACII)*. 2017: 21-27).
39. Rook, L. Mazza, M.C., Lefter, I. & Brazier F.M. - Towards Personalized Linguistic Recognition of Generalized Anxiety Disorder, *Frontiers in Digital Health* (in press).

I Budget table (including the external funding table)

A pluriannual plan will be required if your proposal is accepted

II Letters of intent and letters of commitment

On headed paper of the entity. Letters of commitment are preferable to letters of intent and letters of intent are preferable to the absence of these letters.

III Letter of support from department head of the main lead

On headed paper of the department or institution.

Flagship:	Virtual Reality for Mental Healthcare consortium
Flagship acronym:	Me-Health innovator
Naam Penvoerder:	Prof. dr. Matthias Wieser
Startdatum en looptijd van het project:	01-09-2022 / 01-09-2027
Totaal project budget:	€ 2.000.000,- by Convergence (€ 400.000,- per year) and € 2.000.000,- by matched funding

Levenscenen										2022		2023		2024		2025		2026		2027		Total		Total per partner	
Partner die de kosten maakt										In cash	In kind	In cash	In kind	In cash	In kind	In cash	In kind	In cash	In kind	In cash	In kind	In cash	In kind	In cash	In kind
Onderzoekorganisaties																									
Erasmus MC	positief	0,80	9/1/2022	9/1/2027	Schouw 10	€	4.802		€ 263.630		€ 17.875,32		€ 62.725,96		€ 62.725,96		€ 62.725,96		€ 62.725,96		€ 35.190,64		€ 263.630,00	€ 0	
Erasmus University	positief	0,80	9/1/2022	9/1/2024	Schouw 10	€	4.802		€ 144.000		€ 72.030,00		€ 72.030,00		€ 72.030,00		€ 72.030,00		€ 72.030,00		€ 144.000,00		€ 144.000,00	€ 0	
Erasmus University	positief	0,80	9/1/2024	9/1/2026	Schouw 10	€	4.802		€ 144.000		€ 72.030,00		€ 72.030,00		€ 72.030,00		€ 72.030,00		€ 72.030,00		€ 144.000,00		€ 144.000,00	€ 0	
Technical University Delft	positief	0,80	9/1/2022	9/1/2024	Schouw 10	€	4.802		€ 144.000		€ 72.030,00		€ 72.030,00		€ 72.030,00		€ 72.030,00		€ 72.030,00		€ 144.000,00		€ 144.000,00	€ 0	
Technical University Delft	positief	0,80	9/1/2024	9/1/2026	Schouw 10	€	4.802		€ 144.000		€ 72.030,00		€ 72.030,00		€ 72.030,00		€ 72.030,00		€ 72.030,00		€ 144.000,00		€ 144.000,00	€ 0	
Technical University Delft	PHD TUD	1,00	9/1/2023	9/1/2026	PHD TUD	€	3.151		€ 227.504		€ 56.980,88		€ 56.980,88		€ 56.980,88		€ 56.980,88		€ 56.980,88		€ 227.504,00		€ 227.504,00	€ 0	
Technical University Delft	PHD TUD	1,00	9/1/2022	9/1/2026	PHD TUD	€	3.151		€ 227.504		€ 56.980,88		€ 56.980,88		€ 56.980,88		€ 56.980,88		€ 56.980,88		€ 227.504,00		€ 227.504,00	€ 0	
Erasmus University	PHD EUR	0,50	9/1/2022	9/1/2026	PHD EUR	€	3.518		€ 159.287		€ 32.321,63		€ 32.321,63		€ 32.321,63		€ 32.321,63		€ 32.321,63		€ 159.287,00		€ 159.287,00	€ 0	
Erasmus MC	PHD EMC	0,50	9/1/2022	9/1/2026	PHD EMC	€	3.518		€ 159.287		€ 32.321,63		€ 32.321,63		€ 32.321,63		€ 32.321,63		€ 32.321,63		€ 159.287,00		€ 159.287,00	€ 0	
Erasmus University	project manager	0,80	9/1/2022	9/1/2027	Schouw 9	€	4.371		€ 239.968		€ 15.967,88		€ 47.993,58		€ 47.993,58		€ 47.993,58		€ 47.993,58		€ 31.965,72		€ 239.968,00	€ 0	
Existing funding																									
TUD Technische M. Weinmann	PHD TUD	1,00	9/1/2022	9/1/2026	PHD TUD	€	3.151		€ 227.504		€ 56.980,88		€ 56.980,88		€ 56.980,88		€ 56.980,88		€ 56.980,88		€ 227.504,00		€ 227.504,00	€ 0	
TUD NEDS EU project Kuehner B. de Schutter	PHD TUD	0,20	9/1/2023	9/1/2024	PHD TUD	€	3.151		€ 11.788,80		€ 5.894,40		€ 5.894,40		€ 5.894,40		€ 5.894,40		€ 5.894,40		€ 11.788,80		€ 11.788,80	€ 0	
TUD ERC advanced grant Cleaveland B. de Schutter	PHD TUD	0,20	9/1/2022	9/1/2026	PHD TUD	€	3.151		€ 46.585		€ 11.396,18		€ 11.396,18		€ 11.396,18		€ 11.396,18		€ 11.396,18		€ 46.585,00		€ 46.585,00	€ 0	
TUD AIJ Jans Leuning (government funding)	PHD TUD	0,20	9/1/2022	9/1/2026	PHD TUD	€	3.151		€ 3.842,10		€ 11.473,70		€ 11.473,70		€ 11.473,70		€ 11.473,70		€ 11.473,70		€ 3.842,10		€ 3.842,10	€ 0	
TUD Netherlands Funding (government funding)	PHD TUD	0,20	9/1/2022	9/1/2026	PHD TUD	€	3.151		€ 3.842,10																

Kosten van materialen en hulpmiddelen:			Kosten		2022		2023		2024		2025		2026		2027		Totaal		Totaal per partner	
					in cash	in kind	in cash	in kind	in cash	in kind	in cash	in kind	in cash	in kind						
Onderzoekorganisaties																				
Partners die de kosten maakt	Material / hulpmiddel																			
Onderzoekorganisaties	Esmaara MC	Consumables positio WP 1-3	€ 19,200	€ 1,858	€ 3,858	€ 3,858	€ 3,858	€ 3,858	€ 3,858	€ 3,858	€ 3,858	€ 2,000	€ 19,200	€ 0			€ 19,200	€ 0	€ 19,200	€ 0
	Esmaara University	Consumables positio WP1-3	€ 19,200	€ 1,857	€ 3,858	€ 3,858	€ 3,858	€ 3,858	€ 3,858	€ 3,858	€ 3,858	€ 2,000	€ 19,200	€ 0			€ 19,200	€ 0	€ 19,200	€ 0
	Technical University Delft	Consumables positio WP1-2	€ 19,200	€ 1,857	€ 3,858	€ 3,858	€ 3,858	€ 3,858	€ 3,858	€ 3,858	€ 3,858	€ 2,000	€ 19,200	€ 0			€ 19,200	€ 0	€ 19,200	€ 0
	Esmaara University	Consumables PHD WP1-3	€ 19,200	€ 1,857	€ 3,858	€ 3,858	€ 3,858	€ 3,858	€ 3,858	€ 3,858	€ 3,858	€ 2,000	€ 19,200	€ 0			€ 19,200	€ 0	€ 19,200	€ 0
	Technical University Delft	Consumables PHD WP1-2	€ 19,200	€ 1,857	€ 3,858	€ 3,858	€ 3,858	€ 3,858	€ 3,858	€ 3,858	€ 3,858	€ 2,000	€ 19,200	€ 0			€ 19,200	€ 0	€ 19,200	€ 0
	Technical University Delft	Consumables PHD WP1-4	€ 19,200	€ 1,857	€ 3,858	€ 3,858	€ 3,858	€ 3,858	€ 3,858	€ 3,858	€ 3,858	€ 2,000	€ 19,200	€ 0			€ 19,200	€ 0	€ 19,200	€ 0
	Esmaara University	VR Scales Esmaara Behavioral Lab	€ 70,000	€ 10,000	€ 57,000	€ 10,000	€ 10,000	€ 10,000	€ 2,000	€ 2,000	€ 2,000	€ 2,000	€ 10,000	€ 57,000	€ 0		€ 70,000	€ 0	€ 70,000	€ 0
	Esmaara University	Consumables positio VR Class VR Class	€ 10,000	€ 2,000	€ 8,000	€ 2,000	€ 2,000	€ 2,000	€ 2,000	€ 2,000	€ 2,000	€ 2,000	€ 10,000	€ 8,000	€ 0		€ 10,000	€ 0	€ 10,000	€ 0
	Ondernemingen																			
	Advance	Expert time	€ 10,000	€ 2,000	€ 8,000	2000	€ 2,000	2000	€ 2,000	2000	€ 2,000	2000	€ 2,000	€ 10,000	€ 8,000	€ 0		€ 10,000	€ 0	€ 10,000
InvivoIC	Design hours	€ 6,000	€ 1,000	€ 5,000	1200	€ 1,000	1200	€ 1,000	1200	€ 1,000	1200	€ 6,000	€ 5,000	€ 0			€ 6,000	€ 0	€ 6,000	€ 0
PHYQ	Clinical advice and education	€ 15,170	€ 3,034	€ 12,136	3034	€ 3,034	3034	€ 3,034	3034	€ 3,034	3034	€ 15,170	€ 12,136	€ 0			€ 15,170	€ 0	€ 15,170	€ 0
Phenel	Product design services	€ 15,000	€ 3,000	€ 12,000	3000	€ 3,000	3000	€ 3,000	3000	€ 3,000	3000	€ 15,000	€ 12,000	€ 0			€ 15,000	€ 0	€ 15,000	€ 0
SynGene Medical	Consultancy hours and platform licenses	€ 60,000	€ 10,000	€ 50,000	10000	€ 10,000	10000	€ 10,000	10000	€ 10,000	10000	€ 60,000	€ 50,000	€ 0			€ 60,000	€ 0	€ 60,000	€ 0
Advanza Nederland	Advice on patient involvement, dissemination, help with inclusion of participants	€ 12,000	€ 2,000	€ 10,000	400	€ 2,000	400	€ 2,000	400	€ 2,000	400	€ 12,000	€ 10,000	€ 2,000			€ 12,000	€ 0	€ 12,000	€ 0
Esmaara University	Expert time	€ 60,000	€ 12,000	€ 48,000	12000	€ 12,000	12000	€ 12,000	12000	€ 12,000	12000	€ 60,000	€ 48,000	€ 0			€ 60,000	€ 0	€ 60,000	€ 0
Phylaris	Consulting hours	€ 40,000	€ 8,000	€ 32,000	8000	€ 8,000	8000	€ 8,000	8000	€ 8,000	8000	€ 40,000	€ 32,000	€ 0			€ 40,000	€ 0	€ 40,000	€ 0
PhylisAnalytics	Sensor technologies and data dashboarding	€ 10,000	€ 2,000	€ 8,000	2000	€ 2,000	2000	€ 2,000	2000	€ 2,000	2000	€ 10,000	€ 8,000	€ 0			€ 10,000	€ 0	€ 10,000	€ 0
Overleg (private) partners																				
Team 1																	€ 0	€ 0	€ 0	€ 0
Team 2																	€ 0	€ 0	€ 0	€ 0
Team 3																	€ 0	€ 0	€ 0	€ 0
Team 4																	€ 0	€ 0	€ 0	€ 0
Totaal			€ 413,910														Totaal kosten materialen en hulpmiddelen	€ 413,910		€ 0,00

[illegible]

[illegible]

FINANCIERING - Bijdragen per partner per jaar																																					
		2022				2023				2024				2025				2026				2027				Totaal		Totaal per partner									
		In Euro																																			
		In cash	In kind	In cash	In kind	In cash	In kind	In cash	In kind	In cash	In kind	In cash	In kind	In cash	In kind	In cash	In kind	In cash	In kind	In cash	In kind	In cash	In kind														
Onderzoeksgroepen	Erkijting funding																																				
	TUD Teneuzestraat M. Wiersma	€ 56.980,88	€			€ 56.980,88	€			€ 56.980,88	€			€ 56.980,88	€								€ 227.524,80	€		€ 227.524,80											
	TUD HODD EU project Saatchi B. de Schutter	€ 11.703,80	€			€ 5.851,90	€			€ 11.703,80	€			€ 11.703,80	€								€ 117.038,00	€		€ 117.038,00											
	TUD ERC advanced grant Clariëtte B. de Schutter	€ 11.366,18	€			€ 11.366,18	€			€ 11.366,18	€			€ 11.366,18	€								€ 45.085,08	€		€ 45.085,08											
	TUD EU Adv. funding (government funding)	€ 11.473,70	€			€ 11.473,70	€			€ 11.473,70	€			€ 11.473,70	€								€ 34.421,00	€		€ 34.421,00											
	TUD EU call HORIZON-CG-2021 TRANSFORMATION-01 (R. Marquavee, E. Elemenens)	€ 11.473,70	€			€ 11.473,70	€			€ 11.473,70	€			€ 11.473,70	€								€ 34.421,00	€		€ 34.421,00											
	TUD EU call HORIZON-CG-2021 TRANSFORMATION-01 (R. Marquavee, E. Elemenens)	€ 28.466,45	€			€ 28.466,45	€			€ 28.466,45	€			€ 28.466,45	€								€ 113.862,00	€		€ 113.862,00											
	TUD Bioaction funding (government funding)	€ 25.830,80	€																				€ 25.830,80	€		€ 25.830,80											
	EMC Nextcell 1-year grant (collab. with Paris) J.M. Pijpe	€ 32.321	€			€ 32.321	€			€ 32.321	€			€ 32.321	€								€ 65.643,00	€		€ 65.643,00											
	EMC Confine-NE21 grant J.M. Pijpe/J.M. Seelaar	€ 32.322	€			€ 32.322	€			€ 32.322	€			€ 32.322	€								€ 129.287,00	€		€ 129.287,00											
Onderzoeksgroepen	EMC-PM2 GENP-grant grant R. Seelaar	€ 18.368	€			€ 18.368	€			€ 18.368	€			€ 18.368	€								€ 36.735,00	€		€ 36.735,00											
	EMC-PM2 GENP-grant grant R. Seelaar	€ 18.368	€			€ 18.368	€			€ 18.368	€			€ 18.368	€								€ 36.735,00	€		€ 36.735,00											
	EMC Confine-NE21 grant J.M. Pijpe/J.M. Seelaar	€ 32.326	€			€ 32.326	€			€ 32.326	€			€ 32.326	€								€ 141.178,00	€		€ 141.178,00											
	Future funding																																				
	EMC Next J.M. Pijpe	€ 45.019	€			€ 45.019	€			€ 45.019	€			€ 45.019	€								€ 90.038,00	€		€ 90.038,00											
	EMC Next J.M. Pijpe	€ 45.019	€			€ 45.019	€			€ 45.019	€			€ 45.019	€								€ 90.038,00	€		€ 90.038,00											
	TUD Nici M. Wiersma	€ 4.120,29	€			€ 4.120,29	€			€ 4.120,29	€			€ 4.120,29	€								€ 165.144,00	€		€ 165.144,00											
	TUD Nici M. Wiersma	€ 4.120,29	€			€ 4.120,29	€			€ 4.120,29	€			€ 4.120,29	€								€ 165.144,00	€		€ 165.144,00											
	EUR/MO Open Constitute J.M. Wisseer	€ 32.322	€			€ 32.322	€			€ 32.322	€			€ 32.322	€								€ 129.287,00	€		€ 129.287,00											
	EUR/MO Open Constitute J.M. Wisseer	€ 32.322	€			€ 32.322	€			€ 32.322	€			€ 32.322	€								€ 129.287,00	€		€ 129.287,00											
Onderzoeksgroepen	EUR Vival Cerebra M.J. Wisseer	€ 63.026	€			€ 63.026	€			€ 63.026	€			€ 63.026	€								€ 126.053,00	€		€ 126.053,00											
	EUR Vival Cerebra M.J. Wisseer	€ 63.026	€			€ 63.026	€			€ 63.026	€			€ 63.026	€								€ 126.053,00	€		€ 126.053,00											
	VR Streeklab Erasmus Behaaiel Lab	€ 2.000,00	€	€ 57.000,00	€	€ 13.000,00	€			€ 2.000,00	€			€ 2.000,00	€								€ 13.000,00	€	€ 57.000,00	€ 70.000,00											
	€ 2.000,00	€	€ 57.000,00	€	€ 13.000,00	€			€ 2.000,00	€			€ 2.000,00	€									€ 13.000,00	€	€ 57.000,00	€ 70.000,00											
	Multi-mode multi-processor computing cluster	€ 200,00	€			€ 200,00	€			€ 200,00	€			€ 200,00	€								€ 1.000,00	€		€ 1.000,00											
	€ 200,00	€			€ 200,00	€			€ 200,00	€			€ 200,00	€									€ 1.000,00	€		€ 1.000,00											
	Total bijdrage onderzoeksgroepen	€ 402.186	€ 57.000	€ 467.800	€ 467.800	€ 402.186	€ 57.000	€ 467.800	€ 467.800	€ 402.186	€ 57.000	€ 467.800	€ 467.800	€ 402.186	€ 57.000	€ 467.800	€ 467.800	€ 402.186	€ 57.000	€ 467.800	€ 467.800	€ 402.186	€ 57.000	€ 467.800	€ 467.800	€ 402.186	€ 57.000	€ 467.800									
	Ondernemingen																																				
	Almende	€ 2.000	€			€ 2.000	€			€ 2.000	€			€ 2.000	€								€ 0	€	€ 10.000	€	€ 10.000										
	Almende	€ 2.000	€			€ 2.000	€			€ 2.000	€			€ 2.000	€								€ 0	€	€ 10.000	€	€ 10.000										
Almende	€ 2.000	€			€ 2.000	€			€ 2.000	€			€ 2.000	€								€ 0	€	€ 10.000	€	€ 10.000											
Almende	€ 2.000	€			€ 2.000	€			€ 2.000	€			€ 2.000	€								€ 0	€	€ 10.000	€	€ 10.000											
Almende	€ 2.000	€			€ 2.000	€			€ 2.000	€			€ 2.000	€								€ 0	€	€ 10.000	€	€ 10.000											
Almende	€ 2.000	€			€ 2.000	€			€ 2.000	€			€ 2.000	€								€ 0	€	€ 10.000	€	€ 10.000											
Almende	€ 2.000	€			€ 2.000	€			€ 2.000	€			€ 2.000	€								€ 0	€	€ 10.000	€	€ 10.000											
Almende	€ 2.000	€			€ 2.000	€			€ 2.000	€			€ 2.000	€								€ 0	€	€ 10.000	€	€ 10.000											
Almende	€ 2.000	€			€ 2.000	€			€ 2.000	€			€ 2.000	€								€ 0	€	€ 10.000	€	€ 10.000											
Almende	€ 2.000	€			€ 2.000	€			€ 2.000	€			€ 2.000	€								€ 0	€	€ 10.000	€	€ 10.000											
Almende	€ 2.000	€			€ 2.000	€			€ 2.000	€			€ 2.000	€								€ 0	€	€ 10.000	€	€ 10.000											
Almende	€ 2.000	€			€ 2.000	€			€ 2.000	€			€ 2.000	€								€ 0	€	€ 10.000	€	€ 10.000											
Almende	€ 2.000	€			€ 2.000	€			€ 2.000	€			€ 2.000	€								€ 0	€	€ 10.000	€	€ 10.000											
Almende	€ 2.000	€			€ 2.000	€			€ 2.000	€			€ 2.000	€								€ 0	€	€ 10.000	€	€ 10.000											
Almende	€ 2.000	€			€ 2.000	€			€ 2.000	€			€ 2.000	€								€ 0	€	€ 10.000	€	€ 10.000											
Almende	€ 2.000	€			€ 2.000	€			€ 2.000	€			€ 2.000	€								€ 0	€	€ 10.000	€	€ 10.000											
Almende	€ 2.000	€			€ 2.000	€			€ 2.000	€			€ 2.000	€								€ 0	€	€ 10.000	€	€ 10.000											
Almende	€ 2.000	€			€ 2.000	€			€ 2.000	€			€ 2.000	€								€ 0	€	€ 10.000	€	€ 10.000											
Almende	€ 2.000	€			€ 2.000	€			€ 2.000	€			€ 2.000	€								€ 0	€	€ 10.000	€	€ 10.000											
Almende	€ 2.000	€			€ 2.000	€			€ 2.000	€			€ 2.000	€								€ 0	€	€ 10.000	€	€ 10.000											
Almende	€ 2.000	€			€ 2.000	€			€ 2.000	€			€ 2.000	€								€ 0	€	€ 10.000	€	€ 10.000											
Almende	€ 2.000	€			€ 2.000	€			€ 2.000	€			€ 2.000	€								€ 0	€	€ 10.000	€	€ 10.000											
Almende	€ 2.000	€			€ 2.000	€			€ 2.000	€			€ 2.000	€								€ 0	€	€ 10.000	€	€ 10.000											
Almende	€ 2.000	€			€ 2.000	€			€ 2.000	€			€ 2.000	€								€ 0	€	€ 10.000	€	€ 10.000											
Almende	€ 2.000	€			€ 2.000	€			€ 2.000	€			€ 2.000	€								€ 0	€	€ 10.000	€	€ 10.000											
Almende	€ 2.000	€			€ 2.000	€			€ 2.000	€			€ 2.000	€								€ 0	€	€ 10.000	€	€ 10.000											
Almende	€ 2.000	€			€ 2.000	€			€ 2.000	€			€ 2.000	€								€ 0	€	€ 10.000	€	€ 10.000											
Almende	€ 2.000	€			€ 2.000	€			€ 2.000	€			€ 2.000	€								€ 0	€	€ 10.000	€	€ 10.000											
Almende	€ 2.000	€			€ 2.000	€			€ 2.000	€			€ 2.000	€								€ 0	€	€ 10.000	€	€ 10.000											
Almende	€ 2.000	€			€ 2.000	€			€ 2.000	€			€ 2.000	€								€ 0	€	€ 10.000	€	€ 10.000											
Almende	€ 2.000	€			€ 2.000	€			€ 2.000	€			€ 2.000	€								€ 0	€	€ 10.000	€	€ 10.000											
Almende	€ 2.000	€			€ 2.000	€			€ 2.000	€			€ 2.000	€								€ 0	€	€ 10.000	€	€ 10.000											
Almende	€ 2.000	€			€ 2.000	€			€ 2.000	€			€ 2.000	€								€ 0	€	€ 10.000	€	€ 10.000											
Almende	€ 2.000	€			€ 2.000	€			€ 2.000	€			€ 2.000	€								€ 0	€	€ 10.000	€	€ 10.000											
Almende	€ 2.000	€			€ 2.000	€			€ 2.000	€			€ 2.000	€								€ 0	€	€ 10.000	€	€ 10.000											
Almende	€ 2.000	€			€ 2.000	€			€ 2.000	€			€ 2.000	€								€ 0</															

To:
Prof. dr. Victor Bekkers
Dean
Erasmus School of Social and Behavioral Sciences (ESSB)
Erasmus University Rotterdam Deans office ESSB, Mandeville (T-)
building, room 15-19|
P.O. Box 1738, 3000 DR Rotterdam |Tel + 31 10 4082065
bekkers@essb.eur.nl

i-psy PsyQ Brijder BV
KvK 27321698
www.brijder.nl
www.psyq.nl
www.i-psy.nl

i-psy PsyQ Brijder is onderdeel
van Parnassia Groep

8 March 2022

LETTER OF INTENT
for the
VR Mental Health Consortium PROJECT

Dear prof. dr. Wieser,

I, Annet Spijker, in my capacity of general manager at PsyQ hereby confirm that PsyQ intends to contribute to the VR Mental Health Consortium project, as applied for by the main applicant, Prof. Dr. Matthias Wieser, full professor at Erasmus University Rotterdam.

PsyQ intends to provide an in-kind contribution of clinical advice and co-creation of virtual reality tools for therapy by a certified clinical psychologist, representing a monetary value of € 15 170 Euro/year and further detailed in the project proposal and budget form.

Yours sincerely,



Name: Annet Spijker

Position: General Manager at PsyQ

Date: 8 March 2022

Letter of Intent for a project partner

Erasmus University Rotterdam
Dean Erasmus School of Social and Behavioral Sciences (ESSB)
attn. Prof. dr. V.J.J.M. Bekkers
P.O. Box 1738
3000 DR Rotterdam

Referentie WE.00/u220038/DB/eb
Datum 14 maart 2022
Behandeld door Dinant Bekkenkamp
Betreft **Letter of Intent - project VR Mental Healthcare Consortium**

Dear prof. dr. Wieser,

I, drs. Marco M. Blom, in my capacity of Scientific Director at Alzheimer Nederland, hereby confirm that Alzheimer Nederland intends to contribute to the VR Mental Healthcare Consortium project, as applied for by the main applicant, Prof. dr. Matthias Wieser, full professor at Erasmus University Rotterdam.

Erasmus Alzheimer Nederland intends to contribute € 10.000 in cash towards the project costs in accordance with the budget in the project proposal and budget form.

Alzheimer Nederland intends to provide an 20-hours in-kind contribution for advise on patient involvement, dissemination through our online channels (e.g. about FTD symptoms and psychosocial dementia-related problems) and help with inclusion of participants, representing a monetary value of € 2000,- and further detailed in the project proposal and budget form.

Yours sincerely,
Alzheimer Nederland



drs. Marco M. Blom
scientific director

Erasmus School of Social and Behavioral Sciences (ESSB)
Attn Prof. dr. Victor Bekkers
P.O. Box 1738
3000 DR, ROTTERDAM
The Netherlands

Technolution B.V.
Burg. Janssingel 1
P.O. Box 2013
2800 BD GOUDA
The Netherlands

T +31 (0)182 59 40 00
E info@technolution.com
I www.technolution.com

Subject: LETTER OF INTENT for the VR Mental Healthcare Consortium PROJECT

Company no. 29031050
ISO 9001 certified

Place: Gouda
Date: March 14, 2022

Dear prof. dr. Bekkers,

I, Serge de Vos, in my capacity of Chief Technology Officer (CTO) at Technolution hereby confirm that Technolution intends to contribute to the VR Mental Healthcare Consortium project, as applied for by the main applicant, Prof. Dr. Matthias Wieser, full professor at Erasmus University Rotterdam.

Technolution is a multidisciplinary company focusing on all aspects of data chains in almost any context, from data acquisition to access and process management. We work on innovative products, systems and technologies for a wide range of clients in various industrial sectors, including mobility, energy, manufacturing, high-tech systems and big science including image and video processing and analysis for healthcare.

Technolution is very interested in the project VR Mental Healthcare Consortium as we foresee that the research performed in this project and the technologies applied in this projects will be highly relevant for us. Therefore, Technolution recommends and supports the project proposal

We are willing to participate in the user group of the project and to actively contribute to the project by providing feedback on the ongoing research, background expertise, giving access to experts at Technolution for consultation, and by giving suggestions for further research.

The above activities would amount to about 80 hours of expert time per year, which would correspond to an in-kind contribution of 80 hours x € 150/hour = € 12.000 per year.

Technolution thus intends to provide an in-kind contribution as detailed above, representing a total monetary value of € 60.000 ex. VAT over a 5 year period.

Yours sincerely,
Technolution B.V.



Serge de Vos
CTO



To:
Prof. dr. Victor Bekkers
Dean
Erasmus School of Social and Behavioral Sciences (ESSB)
Erasmus University Rotterdam Deans office ESSB, Mandeville (T-) building, room 15-19|
P.O. Box 1738, 3000 DR Rotterdam |Tel + 31 10 4082065 | bekkers@essb.eur.nl

2022-03-10

LETTER OF INTENT
for the
VR Mental Healthcare Consortium PROJECT

Dear prof. dr. Wieser,

I, MPE Gijssel, in my capacity of CEO at Kinetic Analysis B.V. hereby confirm that Kinetic Analysis B.V. intends to contribute to the VR Mental Healthcare Consortium project, as applied for by the main applicant, Prof. Dr. Matthias Wieser, full professor at Erasmus University Rotterdam.

Kinetic Analysis B.V. intends to provide an in-kind contribution of providing sensor technologies and data dashboarding for better understanding levels of discomfort and stress in individuals, representing a monetary value of € 10.000 and further detailed in the project proposal and budget form.

Yours sincerely,

Name: Maarten Gijssel

Position: CEO

Date: 10-03-2022

To: Prof. dr. Victor Bekkers

Dean

Erasmus School of Social and Behavioral Sciences (ESSB)

Erasmus University Rotterdam Deans office ESSB, Mandeville (T-) building, room 15-19|

P.O. Box 1738, 3000 DR Rotterdam |Tel + 31 10 4082065 | bekkers@essb.eur.nl

March 10, 2022

LETTER OF SUPPORT
for the
VR Mental Healthcare Consortium PROJECT

Dear prof. dr. Wieser,

I, Hendrik-Jan van der Waal, in my capacity of chair at FTD Lotgenoten hereby confirm that FTD Lotgenoten intends to contribute to the VR Mental Healthcare Consortium project, as applied for by the main applicant, Prof. Dr. Matthias Wieser, full professor at Erasmus University Rotterdam.

FTD Lotgenoten intends to contribute by participating in the end-user advisory board of the VR Mental Healthcare Consortium.

Yours sincerely,



Name: Hendrik-Jan van der Waal

Position: Chair FTD Lotgenoten

Date: March 10, 2022

[Use headed paper of party]

To:

Prof. dr. Victor Bekkers

Dean

Erasmus School of Social and Behavioral Sciences (ESSB)

Erasmus University Rotterdam Deans office ESSB, Mandeville (T-) building, room 15-19|

P.O. Box 1738, 3000 DR Rotterdam | Tel + 31 10 4082065 | bekkers@essb.eur.nl

14-03-2022

LETTER OF INTENT

for the

VR Mental Healthcare Consortium **PROJECT**

Dear prof. dr. Wieser,

I, Mike Verhiel, in my capacity of CEO at Psylaris hereby confirm that Psylaris intends to contribute to the VR Mental Healthcare Consortium project, as applied for by the main applicant, Prof. Dr. Matthias Wieser, full professor at Erasmus University Rotterdam.

Psylaris intends to provide an in-kind contribution of Consulting hours, representing a monetary value of € 40.000 and further detailed in the project proposal and budget form.

Yours sincerely,



Name: Mike Verhiel

Position: CEO

Date: 14-03-2022

Letter of Intent for a project partner



To: Prof. dr. Victor Bekkers

Dean

Erasmus School of Social and Behavioral Sciences (ESSB)

Erasmus University Rotterdam Deans office ESSB, Mandeville (T-) building, room 15-19|

P.O. Box 1738, 3000 DR Rotterdam | Tel + 31 10 4082065 | bekkers@essb.eur.nl

11.03.2022

LETTER OF INTENT
for the
VR Mental Healthcare Consortium PROJECT

Dear prof. dr. Wieser,

I, Dimitra Maria Kontogiorgou, in my capacity of Director at Disrupt BV hereby confirm that Disrupt BV intends to contribute to the VR Mental Healthcare Consortium project, as applied for by the main applicant, Prof. Dr. Matthias Wieser, full professor at Erasmus University Rotterdam.

Disrupt BV intends to provide an in-kind contribution of Product Design Services, representing a monetary value of € 15 000 and further detailed in the project proposal and budget form.

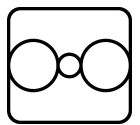
Yours sincerely,

A handwritten signature in blue ink, consisting of several overlapping loops and strokes.

Name: Dimitra Maria

Position: Director

Date: 11.03.2022



To:

Prof. dr. Victor Bekkers

Dean

Erasmus School of Social and Behavioral Sciences (ESSB)

Erasmus University Rotterdam Deans office ESSB, Mandeville (T-) building, room 15-19|

P.O. Box 1738, 3000 DR Rotterdam |Tel + 31 10 4082065 | bekkers@essb.eur.nl

11-03-2022

LETTER OF INTENT
for the
VR Mental Healthcare Consortium PROJECT

Dear prof. dr. Wieser,

I, Robert Brouwer, in my capacity of COO at SyncVR Medical hereby confirm that SyncVR Medical intends to contribute to the VR Mental Healthcare Consortium project, as applied for by the main applicant, Prof. Dr. Matthias Wieser, full professor at Erasmus University Rotterdam.

SyncVR Medical intends to provide an in-kind contribution of consultancy hours and SyncVR platform software licences, representing a monetary value of €50.000 and further detailed in the project proposal and budget form.

Yours sincerely,

Name: Robbert Brouwer

Position: COO and co-founder

Date: 11-03-2022

To: Prof.dr. Matthias Wieser
Erasmus School of Social and Behavioural
Sciences
Erasmus University Rotterdam
Burgemeester Oudlaan 50
3062 PA Rotterdam

Almende BV.
Stationsplein 45
unit d1.116
3013 AK Rotterdam
Nederland

+31 (0)10 404 9444
info@almende.org
www.almende.com

Rotterdam, Tuesday, March 1st

LETTER OF INTENT
for the
PROJECT VR Mental Healthcare Technologies

Dear Prof. Wieser,

I, Aad Nales in my capacity of Partner at Almende B.V. hereby confirm that Almende B.V. intends to contribute to the project *VR Mental Healthcare Technologies*, as applied for by the main applicant, prof.dr. Matthias Wieser, full professor at Erasmus University Rotterdam.

Almende B.V. aims to apply the principles of self-organization to create innovative ICT solutions to empower human beings to better organize their lives in an increasingly complex world.

Almende B.V. is very interested in the project *VR Mental Healthcare Technologies* as we foresee that the research performed in this project and the technologies applied in this project will be highly relevant for us. Therefore, Almende B.V. recommends and supports the project proposal.

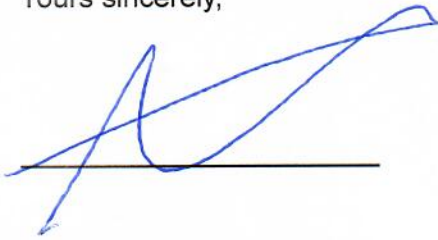
We are willing to participate in the user group of the project and to actively contribute to the project by providing feedback on the ongoing research, background expertise, giving access to experts at Almende for consultation, and by giving suggestions for further research.

The above activities would amount to an estimated 10 hours of expert time per year, which would



correspond to an in-kind contribution of 10 hours x 200 EUR/hour = 2000 EUR per year.

Yours sincerely,

A handwritten signature in blue ink, appearing to be "Aad Nales", written over a horizontal line.

Name: Aad Nales, MSc

Position: Partner

Date: March 1st, 2022

A small, stylized handwritten mark or signature in blue ink at the bottom right of the page.

[Use headed paper of party]

To:

Prof. dr. Victor Bekkers

Dean

Erasmus School of Social and Behavioral Sciences (ESSB)

Erasmus University Rotterdam Deans office ESSB, Mandeville (T-) building, room 15-19|

P.O. Box 1738, 3000 DR Rotterdam |Tel + 31 10 4082065 | bekkers@essb.eur.nl

14-03-2022

LETTER OF INTENT
for the
VR Mental Healthcare Consortium PROJECT

Dear prof. dr. Wieser,

I, Mike Verhiel, in my capacity of CEO at Psylaris hereby confirm that Psylaris intends to contribute to the VR Mental Healthcare Consortium project, as applied for by the main applicant, Prof. Dr. Matthias Wieser, full professor at Erasmus University Rotterdam.

Psylaris intends to provide an in-kind contribution of Consulting hours, representing a monetary value of € 40.000 and further detailed in the project proposal and budget form.

Yours sincerely,



Name: Mike Verhiel

Position: CEO

Date: 14-03-2022

Prof. dr. Victor Bekkers

Dean

Erasmus School of Social and Behavioral Sciences (ESSB)

Erasmus University Rotterdam Deans office ESSB, Mandeville (T-) building, room 15-19| P.O. Box 1738,
3000 DR Rotterdam |Tel + 31 10 4082065 | bekkers@essb.eur.nl

March 14th, 2022

LETTER OF INTENT
for the
VR Mental Health Care Consortium PROJECT

Dear prof. dr. Wieser,

I, Georgia Mitsi, in my capacity of general manager at Biogen Digital Health hereby confirm that Biogen Digital Health intends to contribute to the VR Mental Health Care Consortium project, as applied for by the main applicant, Matthias Wieser, full professor at Erasmus University Rotterdam.

For the avoidance of doubt, this letter of intent does not constitute a legally binding undertaking by Biogen and the principle of the contribution as well as the final amount will have to be approved as per Biogen internal policies.

Yours sincerely,


Name: Georgia Mitsi

Position: General Manager

Date: March 14th, 2022

To whom it may concern

Date

11 March 2022

Subject

Support letter prof. dr. M. Wiesser

Our reference

ESSB/VB/21452/nbo

Your reference

Page

1/1

Appendix

Department

Dean's Office

Visiting address

Erasmus University Rotterdam
Burgemeester Oudlaan 50
Mandeville
T15-19

Postal address

PO Box 1738
3000 DR Rotterdam
The Netherlands

T +31 10 408 2064

E office.dean@essb.eur.nl

W www.eur.nl/essb/English

To whom it may concern

I, Prof. Dr. Victor Bekkers, in my capacity as dean of the Erasmus School of Social and Behavioural Sciences hereby confirm that the faculty intends to fully support the main applicant, Prof. Dr. Matthias Wieser, full professor at Erasmus University Rotterdam, if the proposal "VR Mental Healthcare Consortium", is funded by the convergence initiative.

The host institution will support the main applicant Prof. Dr. Matthias Wieser and his team by

- Hosting the main applicant for the duration of the project
- Providing -during the implementation of the project – research support for the main applicant and team members (regarding infrastructure, equipment, access rights, and other services necessary for conducting the research)
- providing administrative assistance to the main applicant

Yours sincerely,



Prof. dr. V.J.J.M. Bekkers
Dean of the Erasmus School of Social and Behavioural Sciences,